
Counter-Small Unmanned Aircraft System (C-sUAS) Gunnery Program

MAY 2025

DISTRIBUTION RESTRICTION: Approved for public release, distribution is unlimited

HEADQUARTERS, DEPARTMENT OF THE ARMY

This publication is available at the Army Publishing Directorate site (<https://armypubs.army.mil>) and the Central Army Registry Site (<https://atiam.train.army.mil/catalog/dashboard>).

Counter-Small Unmanned Aircraft System (C-sUAS) Gunnery

Contents

	Page
Preface	v
Introduction	vii
CHAPTER 1	1
CERTIFICATION REQUIREMENTS	1
Overview	1
CHAPTER 2	5
LIDS CERTIFICATION AND QUALIFICATION	5
M-LIDS Gunnery	5
FS-LIDS Gunnery	20
CHAPTER 3	27
FIRE CONTROL PROCESS	27
Effective Fire Control	27
Engagement Area.....	28
APPENDIX A	31
PLANNING AND ESTABLISHING RANGES FOR C-SUAS GUNNERY	31
APPENDIX B	37
SCORECARDS	37
APPENDIX C	39
C-SUAS WRITTEN TEST EXAMPLE	39
APPENDIX D	43
EMERGING C-UAS SYSTEMS	43
APPENDIX E	47
HANDHELD CERTIFICATION AND QUALIFICATION	47

Glossary	61
References	63
Index	65

Tables

Table 2-1. M-LIDS Gunnery Tables	5
Table 2-2. Example Crew Table I	6
Table 2-3. GST Task, Conditions, and Standards	6
Table 2-4. LIDS Gunnery Skills Test.....	7
Table 2-5. Example Crew Table II	8
Table 2-6. Table II, Task, Conditions, and Standards	9
Table 2-7. Mobile-low, slow, small unmanned aircraft system integrated defeat system (M-LIDS) Crew Proficiency Table II.....	9
Table 2-8. Example Crew Table III	10
Table 2-9. Table III, Task, Conditions, and Standards	11
Table 2-10. Table III, M-LIDS Crew Proficiency Requirements	11
Table 2-11. Example Crew Table IV	12
Table 2-12. Table IV, Task, Conditions, and Standards	13
Table 2-13. Table IV, Mobile-low, slow, small unmanned aircraft system integrated defeat system (M-LIDS) Crew Proficiency Requirements	14
Table 2-14. Example Crew Table V	14
Table 2-15. Table V, Task, Conditions, and Standards	15
Table 2-16. Table V, M-LIDS Crew Proficiency Requirements.....	15
Table 2-17. Example Crew Table VI	16
Table 2-18. Table VI, Task, Conditions, and Standards	17
Table 2-19. Table VI, Mobile-low, slow, small unmanned aircraft integrated defeat system (M-LIDS) Crew Proficiency Requirements.....	18
Table 2-20. FS-LIDS Crew Roles and Responsibilities	20
Table 2-21. FS-LIDS Gunnery Tables	21
Table 2-22. Table I, Task, Conditions, and Standards	22
Table 2-23. Table II, Task, Conditions, and Standards	22
Table 2-24. Table III, Task, Conditions, and Standards	23
Table 2-25. Table IV, Task, Conditions, and Standards	23
Table 2-26. Table IV, Task, Conditions, and Standards	24
Table 2-27. Table VI, Task, Conditions, and Standards	24
Table 2-28. Fixed Site-Low, Slow, Small Unmanned Aircraft Integrated Defeat System (FS-LIDS) Battle Drills	26
Table B-1. Leaders Counter small Unmanned Aircraft Systems (C-sUAS) Skills Proficiency Test Evaluation Scorecard	37
Table B-2. Operators Counter small Unmanned Aircraft Systems (C-sUAS) Skills Proficiency Test Evaluation Scorecard	37
Table E-1. Counter small Unmanned Aircraft Systems (C-sUAS) Tables I through VI	47
Table E-2. Table I, Resources Required.....	47
Table E-3. Table I, Task, Conditions, and Standards	48
Table E-4. Table I, Test Requirements	48
Table E-5. Table I, Counter small Unmanned Aircraft Systems (C-sUAS) Skills Proficiency Test Certification Requirements Common to All	49
Table E-6. Table II, Resources Required.....	49
Table E-7. Table II, Task, Conditions, and Standards	50
Table E-8. Table II, Individual and Collective Tasks	50
Table E-9. Performance Measures	50
Table E-10. Table II, Threat Identification and Reporting Drills	51

Table E-11. Table III, Resources Required	51
Table E-12. Table III, Task, Conditions, and Standards.....	52
Table E-13. Table III, Individual Tasks	52
Table E-14. Table III, Performance Measures	52
Table E-15. Table III, Individual Systems Calibrate/Zero Equipment.....	53
Table E-16. Table IV, Resources Required.....	53
Table E-17. Table IV, Task, Conditions, and Standards	54
Table E-18. Table IV, Individual Tasks Practice.....	54
Table E-19. Table IV, Performance Measures	54
Table E-20. Table IV, Individual Systems Practice	55
Table E-21. Table V, Resources Required.....	55
Table E-22. Table V, Task, Conditions, and Standards	56
Table E-23. Table V, Individual Qualification Practice	56
Table E-24. Table V, Performance Measures	56
Table E-25. Table V, Individual Qualification	57
Table E-26. Table VI, Resources Required.....	57
Table E-27. Table VI, Task, Conditions, and Standards	58
Table E-28. Table VI, Individual Qualification	58
Table E-29. Table VI, Performance Measures	58
Table E-30. Table VI, Collective Qualification	59

This page intentionally left blank.

Preface

This training circular (TC) standardizes the counter-small unmanned aircraft system (C-sUAS) gunnery program and skill qualifications through performance-oriented, sequential, progressive, realistic, and challenging training. TC 3-01.81 is used to develop and test the proficiency of individuals, crews, and battery personnel in C-sUAS gunnery tasks. This TC references the current crew drills, collective tasks, unit task list, and subtasks to be performed by individuals and crews for gunnery certification.

The target audience for this publication are commanders, trainers, evaluators, leaders, and service members assigned to units at the brigade level and below. It could also aid the joint force commander in planning and executing cohesive joint operations against unmanned aircraft system (UAS) threats throughout the battlefield and can be beneficial for sensor employment and force application.

Commanders, staffs, and subordinates ensure that their decisions and actions comply with applicable United States, international, and in some cases host-nation laws and regulations. Commanders at all levels ensure that their service members operate in accordance with (IAW) the law of war and the rules of engagement. (See FM 6-27 The Commander's Handbook on the Law of Land Warfare).

TC 3-01.81 uses joint terms where applicable. Selected joint and Army terms and definitions appear in the glossary and the text. For other definitions in the text, the term is italicized, and the number of the proponent publication follows the definition.

This TC applies to all US armed services unless otherwise stated.

The proponent for this publication is the United States Army Fires Center of Excellence and Fort Sill. The preparing agency is the Directorate of Training and Doctrine, Air Defense Artillery (ADA) Branch. Send written comments and recommendations on DA Form 2028 (Recommended Changes to Publications and Blank Forms) directly to Commandant, United States Army Air Defense Artillery School, ATTN: ATSF-DD, Fort Sill, OK 73503 or submit an electronic DA Form 2028 by email using the following link: usarmy.sill.fcoe.mbx.dotd-doctrine-inbox@army.mil.

This page intentionally left blank.

Introduction

This proponent publication provides trainers and commanders with gunnery qualification requirements and is a guide to reach published goals and sustain unit and crew proficiency. Training will replicate real world situations and potential future operations, to motivate our service members to execute each task to standard every time. This TC also provides a general overview of C-sUAS unit operations, to assist in the qualification process. Units equipped with C-sUAS capabilities should adhere to the gunnery tables that apply to their specific capabilities. If a unit only has LIDS (fixed-site [FS] or mobile [M]), they only need the gunnery tables for the family system of LIDS (Chapter 2). If a unit only has non-program of record counter-unmanned aircraft system (C-UAS) capabilities, they should reference Appendix D.

Chapters presented within TC 3-01.81 discuss gunnery table qualifications for the C-sUAS platforms. These chapters are:

- **Chapter 1** covers certification requirements.
- **Chapter 2** covers lids certification and qualification.
- **Chapter 3** covers target engagement process.
- **Appendix A** covers planning and establishing gunnery ranges.
- **Appendix B** covers score cards.
- **Appendix C** covers written test example.
- **Appendix D** covers a foundational overview of knowledge needed to operate generalized C-UAS equipment.
- **Appendix E** covers handheld certification and qualification.

This page intentionally left blank.

Chapter 1

Certification Requirements

Crew requirements provide guidance on how to conduct of C-sUAS certification gunnery tables. Training and evaluations must replicate real world situations and potential future operations. Evaluations must be scenarios that exercise the leader's ability to lead and make decisions that enable service members to successfully perform at all levels.

OVERVIEW

1-1. Training tables provide mandatory qualification standards and training strategies for C-sUAS. These training tables focus on preparing the individual to perform as part of a crew to accomplish the unit mission. Standards outlined in the combined arms training strategy (CATS) and service member's training publications, are the minimum acceptable levels of performance.

1-2. The gunnery strategy is based on the building block approach. Individuals are trained in basic skills before being integrated into crews. Crews train progressively from basic tasks, intermediate tasks, and advanced tasks through integration as battery or battalion elements performing their wartime mission.

1-3. Commanders must actively exercise flexibility in applying these strategies, tailoring them to their unit's unique needs and objectives, and they have the discretion to integrate mission command in accordance with ADP 6-0. Commanders can adapt their tactics to respond effectively to emerging situations and make decisions that enable their units to achieve their missions.

1-4. Flexibility in strategy and emphasis on gunnery proficiency empowers unit commanders to make informed decisions, capitalize on opportunities, and mitigate risks, ultimately contributing to the success of their missions.

THREAT

1-5. Unmanned aircraft system (UAS) threats are categorized into groups 1-5 based on weight, size, operating altitude, and operating airspeed. Refer to ATP 3-01.81 to see breakdown of UAS groups. These threats include commercial off the shelf, foreign military, and non-state actors. A UAS is defined as the unmanned aircraft and ground station controller. Threat UAS are used to observe, track, target, and attack friendly forces.

1-6. UAS that target friendly forces generally have low radar signatures, hover, or fly low enough that detection is difficult. Some are controlled by datalinks to a ground controller, while others fly preplanned routes using way points and return to the ground controller without the need for active data links for navigation. UAS that fly preplanned routes using global navigation satellite information are less susceptible to electromagnetic warfare capabilities.

RESPONSIBILITIES

1-7. Leaders and trainers are responsible for conducting gunnery training and qualification according to applicable CATS, technical manuals (TM), field manuals (FM), standard operating procedures (SOP), higher headquarters directives, and guidance from the commander.

1-8. Safety remains a priority during all training events. The crew and evaluators must comply with cautions, warnings, and danger notes. Battalion master gunners must ensure standing operating procedures adhere to range control safety policies, commander safety policies, and C-sUAS safety operations.

EVALUATORS

1-9. The next higher-level command will select evaluators from adjacent units, if possible, to evaluate table qualification. Evaluators are selected from a list of personnel certified by their battalion trainer.

1-10. The evaluator has the discretion to halt evaluations for system failures or safety violations. For system failures, corrective actions must be observed, such as fix or fight criteria or maintenance procedures prior to the evaluator halting an evaluation.

1-11. Evaluators ensure that the execution of the drill complies with all applicable TCs, SOPs, and the C-sUAS gunnery program

DECERTIFICATION

1-12. Units will lose their certification when the following occurs:

- Any unit or crew that exceeds recertification time limits.
- Any one of the equipment crews violating Safety, Caution or Warning during certifications, supporting sustainment crew certifications, or standalone crew-certifications.
- Failure to maintain a minimum of two certified C-sUAS crews. If either crew loses any member, a replacement from a sustainment crew can take that crew member's place.

GUNNERY TRAINING

1-13. Units equipped with C-sUAS platforms are responsible for developing a training program. They ensure training plans and standards adhere to the commander's intent and battalion training strategy. Commanders report training issues and personnel status to the battalion.

1-14. Units request assistance from the battalion master trainers, through the S-3, to ensure they are compliant with the C-sUAS gunnery program. The unit commander provides a battle roster of certified crews with qualification dates to the battalion S-3.

1-15. The gunnery training program will be developed to follow a logical progression. Units conduct training in two phases: individual and collective gunnery.

- Individual gunnery trains service members on individual level skills, qualification, and certification tables I-III.
- Collective: The collective gunnery phase develops table IV-VI.

ESSENTIAL WARFIGHTING SKILLS

1-16. Crews achieve proficiency and critical skills through the following tasks:

- Engaging targets from a stationary position.
- Engaging targets in a simulated chemical, biological, radiological, and nuclear environment.
- Engaging targets at night.
- Detecting, identifying, classifying, and discriminating targets as friendly, neutral or threat.
- Acquiring and engaging targets in an urban, woodland, or desert environment.
- Engaging targets using the appropriate technique for the target type.
- Engaging multiple and successive targets.

1-17. Crews and sections must achieve proficiency in the following skills:

- Engaging targets as a crew or section.
- Maintaining situational awareness and ensuring personnel protection.

UNIT TRAINING MANAGEMENT

1-18. The 8-step training model provides a common framework for guiding commanders as they lead and manage unit training and leader development. Effective unit training results from a sound analysis of the unit's mission and its ability to accomplish that mission. The higher unit's mission, the mission-essential task list (METL), and higher commander's guidance drive the commander's selection of collective tasks on which the unit trains to accomplish mission success.

1-19. Commanders and other leaders plan to develop their subordinate leaders-through training, education, and experience-in the three training domains (institutional, operational, and self-development). Leaders use the 8-step training model to integrate leader development into a unit-training plan.

1-20. The commander is central to determining the few tasks on which C-sUAS tasks the unit must train. Commanders should follow the operations process with the assistance of unit leadership.

1-21. Commanders first plan for training through identifying the collective tasks on which to train. They must identify and sequence training events, identify resources required, and provide the guidance necessary to achieve mission readiness. Commanders will exercise mission command to enable their subordinates to determine how they will achieve their C-sUAS training objectives.

1-22. Thorough preparation ensures that training conditions reflect the expected mission and that commanders have the resources and enablers necessary to train. Commanders execute the training and then assess the training. Assessments aid commanders in determining if units need to retrain tasks and if the training plan requires modification.

COMMANDER'S ASSESSMENT

1-23. Gunnery programs must be well designed and continuous for units to achieve, improve, and sustain gunnery proficiency. Commanders assess and evaluate all aspects of the training plan, preparation, execution, and ultimate task proficiency. Only the unit commander can assess the readiness of a task.

1-24. Assessment refers to the leader's judgment of the organization's ability to perform its tasks and accomplish its mission. Evaluation refers to the process used to measure the demonstrated ability of individuals and units to accomplish specified training objectives and achieve task proficiency. Leaders continuously monitor the unit's METL proficiency and the progress of the unit-training plan.

1-25. The following factors must be considered during the assessment:

- Their own observations and those of subordinate leaders and other individuals.
- Feedback from after action reviews.
- Evaluations of previous training. The commander assesses training with input from the BN executive officer, S-3, subordinate leaders, and unit trainers. They can use individual and collective training to determine the unit's proficiency.
- Individual. Common task test results show basic individual training proficiency. The C-sUAS gunnery skills test evaluates specific platform related individual proficiency. Individual training and qualification scores indicate proficiency levels with the various C-sUAS platforms.
- Collective. Commanders must select collective tasks that support the mission and adapt to unexpected missions.

1-26. During mission analysis the commander should consider the following:

- The unit's current readiness assessment of the collective tasks to train.
- The higher commander's guidance.
- The unit's METL.
- Time available to train.
- The expected operational environment.
- Risks involved in not training collective tasks that the mission might require.
- Any resources needed for training that are not readily available at home station.
- Input from subordinates.

LIVE FIRE TRAINING

1-27. Table V with prescribed equipment requires a passing score as a prerequisite to firing live ammunition.

1-28. These tasks are recommended as minimum requirements driven by the CATS for each proponent. The purpose of these tasks is to ensure the crew possesses the skills and experience necessary to execute live fire gunnery.

This page intentionally left blank.

Chapter 2

LIDS Certification and Qualification

M-LIDS GUNNERY

2-1. Commander, unit trainer, and planners must have a comprehensive understanding of the topics in this chapter, TC 3-20.0, *Integrated Weapons Training Strategy (IWTS)*, and can forecast and plan months in advance to achieve the desired level of proficiency.

2-2. The ammunition requirements (found later in this publication) may be different from the training ammunition authorized within the approved Department of the Army Pamphlet (DA Pam) 350-38, *Standards in Weapons Training*. Commanders, unit trainer, and planners must be able to understand the differences, if any, and request resources accordingly.

2-3. The mobile-low, slow, small unmanned aircraft integrated defeat system (M-LIDS) gunnery program is nested within the Integrated Weapons Training Strategy (known as IWTS). It represents training and qualification at the crew level. It is important for units to understand the basic concepts and principles of the IWTS.

2-4. Table 2-1 shows the crew tables and provides the general description and primary method of training. The environment column indicates a generalized training environment for the event and includes the associated training principle or method for the event.

2-5. Battalion and battery commanders are authorized to choose which engagements are day or night, offensive or defensive, and which engagement will be under chemical, biological, radiological, and nuclear (CBRN) conditions throughout all tables. The battalion standardization officer must brief the commanders on the capabilities of the live fire complex to used so that the commanders can make an informed decision on the engagements.

2-6. The engagement scenarios outlined in Tables II through VI involve electromagnetic attacks and Coyote interceptor engagements. However, if the range constraints do not permit the use of electromagnetic attack weapons alongside lethal defeat weapons, commanders may opt to skip the electromagnetic attack engagements for now. In such cases, they will adhere to the fixed site-low, slow, small unmanned aircraft system integrated defeat system (FS-LIDS) guidelines to complete the electromagnetic attack gunnery tables at a later time.

2-7. Live firing Coyote interceptors is not permitted during training exercises. Instead, all Coyote engagements will be simulated using training aids, devices, simulators, and simulations (TADSS), provided that the system is available.

Table 2-1. M-LIDS Gunnery Tables

PREREQUISITES				LIVE FIRE		
Training Strategy	Table I (Individual)	Table II (Crew)	Table III (Crew)	Table IV (Crew)	Table V (Crew)	Table VI (Crew)
Platoon						
Section						
Crew	Gunnery Skill Test (GST)	STX-V	STX	Basic	Practice	Qualification
Method	Basic	Basic	Intermediate	Intermediate	Advanced	Advanced
STX-situational training exercise STX-V situational training exercise-virtual						

2-8. The following section discusses the specific prerequisites for the crew tables. It includes the task, conditions, and standards crews are required to achieve prior to executing any live fire event, including boresight, and zeroing procedures. It also includes the scoring criteria.

2-9. The live fire prerequisites are specific to tables I, II, and III and will be completed prior to the live fire density. Table III may be conducted during the gunnery density, provided it is successfully completed prior to any live fire event, including boresight, and zeroing.

2-10. Each prerequisite must be completed between T-6 and T-week prior to any boresight, zero, or other live fire event.

TABLE I, GUNNERY SKILLS TEST

2-11. The Crew Table I (table 2-2), Gunnery Skills Test (GST), is a pre-live-fire, hands-on performance, and written evaluation of weaponry tasks, characteristics, capabilities, and function in the live environment. It verifies the service member's ability to perform critical tasks and employ the weapons and systems safely during training and tactical operations.

2-12. The GST evaluates the individual crewmember's ability to execute selected fundamental gunnery-related skills (see table 2-4). Table I evaluates the entire crew's ability to execute selected tasks that are critical to the safe, and successful live fire training. Table I is a mandatory prerequisite for any live fire event, including zero procedures, live fire accuracy screening test, or any live fire maintenance procedure.

Table 2-2. Example Crew Table I

CREW TABLE I	Table I BASIC LIVE	Table II BASIC VIRTUAL	Table III INTERMEDIATE LIVE TADSS	Table IV INTERMEDIATE LIVE FIRE	Table V ADVANCED LIVE FIRE	Table VI ADVANCED LIVE FIRE
CREW PLATFORM	GST	STX-V	STX	BASIC	PRACTICE	QUAL
Note. Table I, II, III can be conducted in any order within the T-6 to T-week timeframe. This provides flexibility when developing the unit training plan. GST-Gunnery Skills Test or gunner's QUAL-qualification STX-situational training exercise STX-V situational training exercise-virtual TADSS-training aids, devices, simulators, and simulations						

2-13. All personnel assigned to a vehicle crew regardless of military occupational specialty who will be participating in any live-fire event will be administered Table I - GST. Commanders will require all service members to conduct the Table I, Gunnery Skills Test tasks from all crew positions. This allows service members to change positions when required during training. This training strategy helps the commander overcome such circumstances as crew changes or unforeseen absences of assigned service members.

2-14. Units must complete Table I, GST, for record, between T-6 and T-execution (T-week) of the live-fire event. Units can include additional tasks they require to support the training program but cannot eliminate tasks for any weapon system. Units should consider including crewmember evacuation, fire drills, and roll-over drills as part of their GST training as well as supporting their unit's driver's training program.

2-15. Training prior to Table I is a platoon responsibility. Table I, GST, is a test administered by a Master Gunner/certified evaluator that serves as the commander's official record of training proficiency prior to entering a live-fire training density. See Table 2-3 for the task, conditions, and standard for Table I and Table 2-4 for Gunnery Skills Test tasks.

Note. Reference Central Army Registry (also known as the CAR) for electromagnetic warfare tasks.

Table 2-3. GST Task, Conditions, and Standards

Table I, Gunnery Skills Test	
Task	Demonstrate proficiency on weapon or platform specific gunnery and safety-related skills.
Conditions	Given fully mission capable equipment, weapons, optics, vehicles for the position assigned, and certified evaluators.
Standard	The individuals must successfully pass GST using the evaluation standards between T-6 and T-week.

Table 2-4. LIDS Gunnery Skills Test

GUNNERY SKILLS TEST				
PHASE	TASK NUMBER	TASK	WHERE	FREQUENCY
INDIVIDUAL	441-14A-1077	Visual Aircraft Recognition	Unit Training Area	Monthly
INDIVIDUAL	091-91F-1009	Maintain M240 Machine Gun	Unit Training Area	Monthly
INDIVIDUAL	171-131-3011	Disassembly of M240 Machine Gun	Unit Training Area	Monthly
INDIVIDUAL	171-131-2009	Assemble M240 Machine Gun	Unit Training Area	Monthly
INDIVIDUAL	071-022-0001	Maintain a Caliber .50 M2 Series Machine Gun	Unit Training Area	Monthly
INDIVIDUAL	171-131-2014	Perform Detailed Assembly of an M2 Flex Machine Gun	Unit Training Area	Monthly
INDIVIDUAL	171-131-3013	Perform Detailed Disassembly of an M2 Flex Machine Gun	Unit Training Area	Monthly
INDIVIDUAL	071-030-0040	Operate the Common Remotely Operated Weapon Station (CROWS)	Unit Training Area	Monthly
INDIVIDUAL	071-030-041	Maintain the CROWS	Unit Training Area	Monthly
INDIVIDUAL	Unit SOP	General Knowledge/PMCS	Unit Training Area	Monthly
CREW	44-CW-0380	Install M240 Machine Gun	Unit Training Area	Quarterly
CREW	44-CW-0371	Load M240 Coaxial Machine Gun	Unit Training Area	Quarterly
CREW	44-CW-0381	Remove M240 Coaxial Machine Gun	Unit Training Area	Quarterly
CREW	071-030-0042	Boresight the CROWS	Unit Training Area	Quarterly
CREW	071-030-0043	Engage targets with CROWS	Unit Training Area	Quarterly
CREW	071-030-0045	Mount 50 Cal. Machine gun on the CROWS	Unit Training Area	Quarterly
CREW	071-030-0046	Remove the M2 .50 Cal. Machine Gun from the CROWS	Unit Training Area	Quarterly
CREW	071-030-0047	Load the M2 .50 Caliber Machine Gun Mounted in the CROWS	Unit Training Area	Quarterly
CREW	071-030-0048	Unload the M2 .50 Caliber Machine Gun Mounted in the CROWS	Unit Training Area	Quarterly
CREW	071-030-0049	Adjust the Timing on an M2 .50 Caliber Machine Gun Mounted in the CROWS	Unit Training Area	Quarterly
CREW	44-CW-0372	Install XM914 30MM Cannon	Unit Training Area	Quarterly
CREW	44-CW-0354	Conduct Boresight 30MM with Boresight Device	Unit Training Area /Local Training Area	Quarterly
CREW	44-CW-D0373	Load XM914 30MM Canon	Unit Training Area	Quarterly
CREW	44-CW-0374	Reload XM914 30MM Cannon	Unit Training Area	Quarterly
CREW	44-CW-0376	Remove XM914 30MM Cannon	Unit Training Area	Quarterly
CREW	44-CW-D0366	Conduct 30MM Canon Misfire Procedures	Unit Training Area	Quarterly
CROWS-Common Remotely Operated Weapon Station LIDS-low, slow, small unmanned aircraft system integrated defeat system PMCS-Preventive maintenance checks and services SOP-Standard operating procedures				

Evaluation Procedures

2-16. An evaluator external to the PLT/BTRY logs information about each crewmember on a roster. The evaluator supplies the crewmember with all required materials and equipment, displaying them according to the guidance listed in this manual. The evaluator uses the performance measures to record the crewmember's

performance. Afterward, the evaluator will inform the crewmember how well they performed on this task and directs them to the next task or to further training. One evaluator can only test one crewmember at a time.

2-17. If a crewmember fails to achieve the standard on the performance measures, the crew receives a NO-GO. When this happens, the evaluator explains to the crewmember what they did wrong and how to correct their mistakes. Then, the evaluator retests the crewmember according to the local SOP.

TABLE II, SIMULATIONS TRAINING EXERCISE VIRTUAL

2-18. Crew Platform, Table II, situational training exercise-virtual (also known as STX-V) is a simulations-based demonstration of performance of employing the fire unit to its primary capabilities. It is the culminating evaluations event of the crew after completing the appropriate simulations training strategy. Table II is a replication (simulation) of the crew's live-fire qualification course. It provides a measured demonstration of performance before any live-fire event for the element.

2-19. Table II's specific event is based upon the element's capabilities and mission. Crew Platform, Table II, STX-V general information is in Table 2-5.

Table 2-5. Example Crew Table II

CREW TABLE II	Table I BASIC LIVE	Table II BASIC VIRTUAL	Table III INTERMEDIATE LIVE TADSS	Table IV INTERMEDIATE LIVE FIRE	Table V ADVANCED LIVE FIRE	Table VI ADVANCED LIVE FIRE
CREW PLATFORM	GST	STX-V	STX	BASIC	PRACTICE	QUAL
 For platforms without simulations systems or availability to them (when training in austere environments), the training publication will provide alternate methods for Table II requirements. Note. Table I, II, III can be conducted in any order within the T-6 to T-week timeframe. This provides flexibility when developing the unit training plan.						
GST-gunnery skills test or gunner's QUAL-qualification STX-situational training exercise STX-V-situational training exercise-virtual TADSS-training aids, devices, simulators, and simulations						

2-20. The purpose of Crew Platform, Table II, STX-V, is to ensure the crew possesses the skills and experience necessary to execute live-fire events and exercises safely, which implies a foundational understanding of the engagement process, fire commands, engagement techniques, and lethality. This is the culminating test of their abilities and skills on their platform in simulation after completing the appropriate simulations training plan.

2-21. Table II, STX-V, is a mandatory prerequisite for any live-fire event, including zero procedures, sub caliber or in-bore training event, or any live-fire maintenance procedure within the crew training strategy. Crews complete Table II between T-6 and T-week to support the training events.

2-22. Successful completion of the prescribed exercise or series of exercises in the simulator is a mandatory prerequisite to firing live ammunition. These exercises evaluate the crew's proficiency in employing its weapon system or platform against common threats, provides engagements under simulated CBRN conditions, enhance, and evaluate crew coordination, evaluate crew drills, and hone the crew's engagement techniques.

2-23. The Vehicle Commander, gunner, and driver assigned to a crew have successfully demonstrated the skills and knowledge required to continue training on the platform through platform qualification. Commanders are confident in their service member's abilities in the position assigned.

2-24. The tasks evaluated in Table II provide the commander a means to certify the crew's basic weapons platform proficiency prior to live-fire training. It is used as a guide for identifying individual and crew weapons platform strengths and weaknesses on specific engagement techniques and employment of the

vehicle's fire control system capabilities and enablers. See Table 2-6 for Table II task, condition, and standards.

Table 2-6. Table II, Task, Conditions, and Standards

<i>Table II, STX-V</i>	
Task	Engage and destroy stationary, moving ground, and aerial targets from a stationary position in a simulated environment.
Conditions	Given the Following conditions, a fully operational training aids, devices, simulators, and simulations (TADSS), certified Vehicle Crew Evaluators (VCE's) and unit trainers. Also, a prescribed gate to live fire exercise which includes an accurate replication of Table VI Qualification.
Standard	The crew must successfully pass situational training exercise-virtual (STX-V) using the evaluation standards between T-6 and T-week.

Evaluation Procedures

2-25. Table II is a simulations-based exercise that replicates the engagements of the Table VI, Qualification. Scenarios will be developed on a M-LIDS TADSS equivalent if available. Training and evaluation of the crew for Table II is the responsibility of their platoon leader and platoon sergeant.

2-26. Engagement times can be reduced at the discretion of the unit trainer or commander. Offensive and defensive engagements are at the discretion of the commander. Table 2-7 specifies the targets recommended for development of the scenario.

Table 2-7. Mobile-low, slow, small unmanned aircraft system integrated defeat system (M-LIDS) Crew Proficiency Table II

<i>ENG #</i>	<i>TARGET TYPE</i>	<i>TYPE OF ENGAGEMENT</i>	<i>TIME EXPOSURE</i>	<i>NUMBER OF ROUNDS</i>
1	Troops in the open	240C COAX (GUNNER)	50 seconds	50
2	Aerial	Electromagnetic Attack	50 seconds	N/A
3	Aerial	30MM (GUNNER)	50 seconds	20
4	Troops in the open	240C COAX (GUNNER)	50 seconds	50
	Moving Truck	.50Cal (GUNNER)	50 seconds	25
5	Troops in the open	240C COAX (GUNNER)	50 seconds	50
6	Aerial	COYOTE	50 seconds	1
7	Pop Up Rotary	30MM (GUNNER)	50 seconds	20
8	Troops in the open	240C COAX (GUNNER)	50 seconds	50
	Moving Light Armor Vehicle	.50Cal (GUNNER)	50 seconds	25
9	Aerial	30MM (GUNNER)	50 seconds	20
10	Moving Truck	.50Cal (GUNNER)	50 seconds	25
	Troops in the open	240C COAX (GUNNER)	50 seconds	50
11	Rocket propelled gun (RPG) Team	240C COAX (GUNNER)	50 seconds	50
	Aerial	30MM (GUNNER)	50 seconds	20

Table 2-7. Mobile-low, slow, small unmanned aircraft system integrated defeat system (M-LIDS) Crew Proficiency Table II (continued)

ENG #	TARGET TYPE	TYPE OF ENGAGEMENT	TIME EXPOSURE	NUMBER OF ROUNDS
12	Pop Up Rotary	.50Cal (GUNNER)	50 seconds	25
	Aerial Incoming	30MM (GUNNER)	50 seconds	20
Note. The engagement scenarios outlined in Tables II through VI involve electromagnetic attacks and Coyote interceptor engagements. However, if the range constraints do not permit the use of electromagnetic attack weapons alongside lethal defeat weapons, commanders may opt to skip the electromagnetic attack engagements for now. In such cases, they will adhere to the Fixed site-low, slow, small unmanned aircraft system integrated defeat system (FS-LIDS) guidelines to complete the electromagnetic attack gunnery tables at a later time. Live firing Coyote interceptors is not permitted during training exercises. Instead, all Coyote engagements will be simulated using training aids, devices, simulators, and simulations also known as training aids, devices, simulators, and simulations (TADSS), provided that the system is available.				

TABLE III, STX PROFICIENCY

2-27. Crew Platform, Table III (table 2-8), STX, is a live environment, hands-on training event utilizing training aids and devices, and includes the use of blank ammunition, pyrotechnics, and battlefield effects simulators. This event trains and evaluates the crew's ability to execute critical tasks using their organic weapons, systems, and equipment, during day and limited visibility conditions, and while operating in a CBRN environment.

2-28. Each element must perform tasks and skills to a directed sequence or time standard as listed in the crew platform's training publication. Training and evaluation of the element is the responsibility of their platoon leader and platoon sergeant.

Table 2-8. Example Crew Table III

CREW TABLE III	Table I BASIC LIVE	Table II BASIC VIRTUAL	Table III INTERMEDIATE LIVE TADSS	Table IV INTERMEDIATE LIVE FIRE	Table V ADVANCED LIVE FIRE	Table VI ADVANCED LIVE FIRE
CREW PLATFORM	GST	STX-V	STX	BASIC	PRACTICE	QUAL
 Crew training in an austere environment may not have access to the required training devices to meet the conditions of crew Table III. Units operating under these conditions are provided alternate methods to complete their Table III requirements within the vehicle's training publication. Note. Tables I, II and III can be conducted in any order within the T-6 to T-Week timeframe. This provides flexibility when developing the unit training plan. GST-gunnery skills test or gunner's QUAL-qualification STX-situational training exercise STX-V situational training exercise-virtual TADSS-training aids, devices, simulators, and simulations						

2-29. Each element must perform the tasks and skills to a directed sequence or time standard as listed in the crew platform's training publication. Training and evaluation of the crew is the responsibility of their platoon leader and platoon sergeant.

2-30. Each crew will conduct engagements on three ground targets with the M240, .50 Cal, or 30mm if operating at a live fire exercise Range. All engagements will be simulated if operating in the Unit Training Area/using the Conduct of Fire Trainer or M-LIDS equivalent.

2-31. Table III, Proficiency, is performed on the crew's assigned vehicle. It is designed to train and evaluate the crew's ability to engage stationary and/or moving targets placed in a tactical array, during day and limited visibility conditions and CBRN environment. Crews execute these tasks from the offense i.e. (shooting from the halt) and defense (behind a berm) posture. See Table 2-9 for Table III task, condition, and standards.

Table 2-9. Table III, Task, Conditions, and Standards

Table III, STX	
Task	Engage and destroy stationary, moving, and aerial targets placed in a tactical array, during day and limited visibility from an offensive/defensive position and requires the use of laser-based training aids, devices, simulators, and simulations (TADSS) for all crews.
Conditions	Given the following conditions a fully operational platform with weapon or weapon system, appropriate laser-based training devices, authorized blank ammunition and pyrotechnics, a certified Vehicle Crew Evaluator, and full scale or scaled targets to meet scenario requirements.
Standard	The crew must score a minimum of 700 of 1000 points overall, score 70 points or more on all targets presented on seven out of ten engagements, qualify at least one night engagement, and successfully pass using evaluation standards between T-6 and T-week. Standards are met when crew members have successfully received a GO on all subtasks listed below.

Evaluation Procedures

2-32. Table III validates the crew's ability to complete basic crew engagements using various platform capabilities, provides for day and night driver's training as part of the unit driver's training program, and ensures the crew demonstrates proficiency performing specific emergency procedures on their platform. It provides time for training and evaluation of these skills that are the foundation to all maneuver training focused on the ability to shoot, move, and communicate.

2-33. All engagements are evaluated using certified vehicle crew evaluators using DA Form 8265 (*Common Crew Score Sheet*) and DA Form 8265-1 (*Common Crew Roll-Up*) as the official record of training. Unit trainers must ensure the training records are included in each crew's gunnery packet. Table 2-10 specifies the targets recommended for the development of the scenario.

Firing Group

2-34. Each crew will utilize blanks for engagements on three ground targets with the M240 if operating at an LFX Range. All engagements will be simulated if operating in the Unit Training Area/Simulated Training Environment.

2-35. The engagements may be full-scale or scaled, depending on the usable area within the training area. Crew's target acquisition skills and engagement techniques, using $\frac{3}{4}$ scale targets, is challenged. See TC 25-8, Training Ranges, for additional information on target dimensions for scaled ranges.

Table 2-10. Table III, M-LIDS Crew Proficiency Requirements

ENG#	Target Type	Type of Engagement	Time Exposure	Number of Rounds
1	Aerial	30MM (GUNNER)	50 seconds	20
2	Troops in the open	240C COAX (GUNNER)	50 seconds	50
3	Aerial	COYOTE	50 seconds	1
4	Troops in the open	240C COAX (GUNNER)	50 seconds	50
5	Moving Vehicle	.50Cal (GUNNER)	50 seconds	25
6	Aerial	Electromagnetic Attack	50 seconds	N/A
7	Aerial	30mm (GUNNER)	50 seconds	20
8	Pop Up Rotary/Aerial	240C COAX 30mm (GUNNER)	50 seconds	50 20
9	Troops in the open	240C COAX (GUNNER)	50 seconds	50
10	Pop Up Rotary	.50Cal (GUNNER)	50 seconds	25
11	Aerial	Electromagnetic Attack	50 seconds	N/A
12	Aerial	.50Cal (GUNNER)	50 seconds	25

Table 2-10. Table III, M-LIDS Crew Proficiency Requirements (continued)

ENG#	Target Type	Type of Engagement	Time Exposure	Number of Rounds
13	Aerial	30mm (GUNNER)	50 seconds	20
Note. The engagement scenarios outlined in Tables II through VI involve electromagnetic attacks and Coyote interceptor engagements. However, if the range constraints do not permit the use of electromagnetic attack weapons alongside lethal defeat weapons, commanders may opt to skip the electromagnetic attack engagements for now. In such cases, they will adhere to the FS-LIDS guidelines to complete the electromagnetic attack gunnery tables at a later time. Live firing Coyote interceptors is not permitted during training exercises. Instead, all Coyote engagements will be simulated using training aids, devices, simulators, and simulations also known as TADSS, provided that the system is available. FS-LIDS-fixed site-low, slow, small unmanned aircraft system integrated defeat system				

2-36. All engagement times can be reduced or increased at the Master Evaluator/Commanders discretion. Targets must be fully exposed and locked out before engagement time starts. The commander has the discretion on the number of offensive and defensive engagements.

TABLE IV BASIC

2-37. Crew Platform, Table IV (table 2-11), Basic, is a course of fire using full-caliber training ammunition on an authorized live-fire facility. This table trains basic skills at a reduced tempo to enable proper leader and service member execution of crew skills. This table builds upon those skills trained and tested during Tables I, II, and III.

2-38. It includes a demonstration of the performance of critical skills in a live environment under live-fire conditions. Table IV is commonly the first training event where the element employs its weapons with full caliber training ammunition against prescribed targets. It uses combat-realistic scenarios to a required performance measure and standard, and includes engagements during day, night, limited visibility, and CBRN conditions on demand.

Table 2-11. Example Crew Table IV

CREW TABLE IV	Table I BASIC LIVE	Table II BASIC VIRTUAL	Table III INTERMEDIATE LIVE TADSS	Table IV INTERMEDIATE LIVE FIRE	Table V ADVANCED LIVE FIRE	Table VI ADVANCED LIVE FIRE
CREW PLATFORM	GST	STX-V	STX	BASIC	PRACTICE	QUAL
 Platforms that require alignment, zeroing, confirmation of point of aim (PoA) / point of impact (Pol) or other critical task must complete Table IV prior to conducting any other live fire event with the weapons.						
GST-gunnery skills test or gunner's QUAL-qualification STX-situational training exercise STX-V situational training exercise-virtual TADSS-training aids, devices, simulators, and simulations						

2-39. This table is executed only after any zeroing or boresight procedures have been completed successfully. Crews may execute this table using TADSS only when ammunition is unavailable or for retraining after the live fire training was completed. Training of the crew is the responsibility of their platoon leader and platoon sergeant and the evaluation is the responsibility of the master gunner/unit trainer.

2-40. The design of Table IV is to train and evaluate the crew's ability to engage single and multiple stationary or moving targets placed in a tactical array, during the day, with an operational or simulated degraded fire control system, as applicable. Crews execute these tasks from the defense using full-caliber ammunition for zero and machine gun ammunition for tactical engagements.

2-41. Table IV provides crews with the ability to align their fire control system, screen or zero their ammunition by type, and complete basic machine gun engagement skills from a defensive firing position. There are two task sets to Table IV, Basic:

- Zeroing (30mm, .50Cal, and 7.62mm).
- Defensive posture engagements.

2-42. Required TADSS are SP-1 BORESIGHT Panel and ST-5 Zero Panel, see TC 25-8, Training Ranges, for construction dimensions if not available through the installation TSC. See Table 2-12 for Table IV task, conditions, and standards.

Table 2-12. Table IV, Task, Conditions, and Standards

Table IV, BASIC	
Task	Engage and destroy stationary and moving targets placed in a tactical array, during day using day and thermal channels from a defensive firing position using 30mm, .50Cal, and 7.62mm ammunition.
Conditions	Given the following conditions a fully operational platform with weapon or weapon system, appropriate caliber ammunition, appropriate laser-based training devices, if required, authorized blank ammunition and pyrotechnics, a certified Vehicle Crew Evaluator and a full-scale or scaled targets to meet the scenario requirements.
Standard	The crew must score a minimum of 700 of 1000 points overall. Score 70 points or more on all targets presented on seven out of ten engagements. Qualify at least one night engagement. Standards are met when crew members have successfully received a Passing Score on all subtasks listed below.

2-43. The Vehicle Commander and gunner assigned as a crew have successfully demonstrated the skills and knowledge required to continue training on the platform through platform qualification. Commanders are confident in their service member's abilities in the position assigned.

2-44. The tasks evaluated in Table IV provide the commander a means to verify the crew's basic weapons platform proficiency prior to further live-fire training. It is a guide for the commander to identify individual and crew weapons platform strengths and weaknesses on specific engagement techniques and employment of the vehicle's fire control system capabilities and enablers. The commander and unit trainer uses the results from Basic - Table IV to structure the unit's crew training program.

Defensive Engagements

2-45. Completion of zeroing and bore sighting procedures must occur successfully before conducting Table V, Practice. Vehicles that fail to zero may not progress to any live fire table including the defensive engagements.

2-46. Crews should perform Table IV, Basic, Defensive Engagements, on the crew's assigned vehicle. It is designed to train and evaluate the crew's ability to engage single and multiple stationary and moving targets placed in a tactical array, during the day and night, with a fully operational or simulated degraded fire control system, as depicted in the task, conditions, and standard for each engagement. Crews execute these tasks from the defense using 7.62mm, .50Cal, and 30mm cannon.

2-47. Units should use 3/4 scale targets at full range for all scenarios but may consider using full-scale based on the unit's level of expertise. The unit trainer may consider using F-type silhouettes rather than E-type silhouettes within the RPG and TROOP target sets. Refer to TC 25-8 for more information.

2-48. Crews may execute this table using TADSS only when ammunition is unavailable or to conduct retraining after live fire training is complete.

2-49. If any individual target score on an engagement is less than 70 points, the engagement is unqualified, regardless of overall engagement score. See table 2-13 for M-LIDS on page 14, crew proficiency requirements.

Table 2-13. Table IV, Mobile-low, slow, small unmanned aircraft system integrated defeat system (M-LIDS) Crew Proficiency Requirements

<i>ENG#</i>	<i>Target Type</i>	<i>Type of Engagement</i>	<i>Time Exposure</i>	<i>Number of Rounds</i>
1	Zero	30mm (GUNNER)	N/A	15
2	Zero	.50Cal	N/A	14
3	Zero	240C COAX (GUNNER)	N/A	50

TABLE V, PRACTICE

2-50. Crew Platform, Table V (table 2-14), Practice, is a live environment training event under live-fire conditions using full-caliber training ammunition, realistic targets, and engagement scenarios at a combat tempo. It is conducted on an authorized live-fire facility. Regardless of crew platform type, this table reinforces the basic skills trained during previous events but increases the tempo to expected engagement sequence capabilities to build leader and element tactical and technical mastery.

2-51. This table includes execution of critical skills in a live environment. Table V prepares the crew to employ their weapons and systems against most common threat scenarios using prescribed targets to a required standard.

Table 2-14. Example Crew Table V

<i>CREW TABLE V</i>	<i>Table I BASIC LIVE</i>	<i>Table II BASIC VIRTUAL</i>	<i>Table III INTERMEDIATE LIVE TADSS</i>	<i>Table IV INTERMEDIATE LIVE FIRE</i>	<i>Table V ADVANCED LIVE FIRE</i>	<i>Table VI ADVANCED LIVE FIRE</i>
<i>CREW PLATFORM</i>	<i>GST</i>	<i>STX-V</i>	<i>STX</i>	<i>BASIC</i>	<i>PRACTICE</i>	<i>QUAL</i>
 This event is not required; however, the majority of high payoff skills, engagement techniques, firing positions, and other critical tactical employment task are trained and reinforced during this course of fire. Units that fail to include Table V in their unit training plan dramatically reduce the crew's ability to qualify their platform on the first attempt and reduce the unit's lethality. In general, crew platforms have a minimum of 2,400 basic engagement possibilities, the crews through repetition will achieve proficiency. The training strategy leverages simulations to provide those engagement experiences to each crew. Select high-payoff engagements that demonstrate performance on critical engagement skills are conducted during crew live-fire training GST-gunnery skills test or gunner's QUAL-qualification STX-situational training exercise STX-V situational training exercise-virtual TADSS-training aids, devices, simulators, and simulations						

2-52. This table prepares crews for the Table VI, Qualification. It is designed to be more difficult to master than the qualification. Units should retrain as necessary with full-caliber ammunition when possible. This table includes execution of critical skills in a live environment. Table V prepares the crew to employ their weapons and systems against most common threat scenarios using prescribed targets to a required standard. Training of the crew is the responsibility of their platoon leader and platoon sergeant and the evaluation is the responsibility of the master gunner/unit trainer.

2-53. Table V, Practice, trains and evaluates the crew's ability to engage stationary or moving targets placed in a tactical array, during day and limited visibility conditions, with an operational or simulated degraded fire control system, as applicable. Crews execute these tasks from the offense and defense using full-caliber ammunition against $\frac{3}{4}$ scale targets placed at full range.

2-54. The Vehicle Commander and gunner assigned as a crew have successfully demonstrated the skills and knowledge required to continue training on the platform through platform qualification. Commanders are confident in their service member's abilities in the position assigned.

Evaluation Procedures

2-55. The tasks evaluated in Table V provide the commander with a means to certify the crew's basic and advanced weapons platform proficiency prior to the qualification course of fire using full-caliber munitions. The results of table V are utilized by the Commander and Master Gunner to inform and structure the unit's crew training program, and the outcome is also reported as a key readiness metric.

2-56. Units should plan for at least 3 training days per Battery allowing for range set-up maintenance and mechanical failure.

2-57. Unit trainers must be familiar with the family of maneuver ranges, the range operations control area (ROCA), and the specific instrumentation and target sets available on the range facility selected for Table V.

2-58. Table V will include 14 engagement scenarios with single and multiple targets. Engagements will incrementally increase in complexity. See Table 2-15 for Table V task, conditions, and standards.

Table 2-15. Table V, Task, Conditions, and Standards

<i>Table V, PRACTICE</i>	
Task	Engage and destroy stationary and moving ground and aerial targets from a stationary, short halt (less than 30 seconds), or on-the-move, and degraded positions in a live fire environment.
Conditions	Giving the following conditions a fully operational platform with weapon or weapon system, allotted ammunition, a certified vehicle crew evaluator, and a full-scale or scaled targets to meet the scenario requirements.
Standard	The crew must score a minimum of 700 of 1000 points overall, score 70 points or more on all targets presented on seven out of ten engagements and qualify at least one night engagement. Standards are met when crew members have successfully received a GO on all subtasks listed below.

2-59. If any individual target score on an engagement is less than 70 points, the engagement is 1570 unqualified, regardless of overall engagement score. All engagement times can be reduced or increased at the Master Evaluator/Commanders discretion. Targets must be fully exposed and locked out before engagement time starts. See Table 2-16 for M-LIDS crew proficiency requirements for Table V.

Table 2-16. Table V, M-LIDS Crew Proficiency Requirements

<i>ENG #</i>	<i>Target Type</i>	<i>Type of Engagement</i>	<i>Time Exposure</i>	<i>Day/Night</i>	<i>Number of Rounds</i>
1	Aerial	Electromagnetic Attack	50 seconds	D/N	N/A
2	Troops in the open	240C COAX (GUNNER)	50 seconds	D/N	50
3	Aerial	.50Cal (GUNNER)	50 seconds	D/N	25
4	Aerial	30MM (GUNNER)	50 seconds	D/N	20
5	Troops in the open	240C COAX (GUNNER)	50 seconds	D/N	50
	Moving Truck	.50Cal (GUNNER)	50 seconds	D/N	25
6	Aerial	COYOTE	50 seconds	D/N	1
7	Pop Up Rotary	.50Cal (GUNNER)	50 seconds	D/N	25
	Aerial	30MM (GUNNER)	50 seconds	D/N	20
8	Troops in the open	240C COAX (GUNNER)	50 seconds	D/N	50
9	Aerial	Electromagnetic Attack	50 seconds	D/N	N/A
10	Aerial	30MM (GUNNER)	50 seconds	D/N	20

Table 2-16. Table V, M-LIDS Crew Proficiency Requirements (continued)

ENG #	Target Type	Type of Engagement	Time Exposure	Day/Night	Number of Rounds
11	Aerial	30MM (GUNNER)	50 seconds	D/N	20
	Pop Up Rotary	.50Cal (GUNNER)	50 seconds	D/N	25
12	Rocket propelled gun (RPG) Team	240C COAX (GUNNER)	50 seconds	D/N	50
	Light Truck FRT	.50Cal (GUNNER)	50 seconds	D/N	25
13	Aerial	Electromagnetic Attack	50 seconds	D/N	N/A
14	Pop Up Rotary	.50Cal (GUNNER)	50 seconds	D/N	25
	Aerial	30MM (GUNNER)	50 seconds	D/N	20
Note. The engagement scenarios outlined in Tables II through VI involve electromagnetic attacks and Coyote interceptor engagements. However, if the range constraints do not permit the use of electromagnetic attack weapons alongside lethal defeat weapons, commanders may opt to skip the electromagnetic attack engagements for now. In such cases, they will adhere to the FS-LIDS guidelines to complete the electromagnetic attack gunnery tables at a later time. Live firing Coyote interceptors is not permitted during training exercises. Instead, all Coyote engagements will be simulated using training aids, devices, simulators, and simulations also known as TADSS, provided that the system is available. M-LIDS - mobile-low, slow, small unmanned aircraft integrated defeat system FS-LIDS – fixed site-low, slow, small unmanned aircraft integrated defeat system					

TABLE VI, QUALIFICATION

2-60. Crew Platform, Table VI (table 2-17), Qualification, is the record course of fire used to determine the live-fire proficiency of a crew platform. Table VI serves as the live-fire proficiency gate (LFPG), which is the live fire prerequisite for the crew to move to collective live fire events. Table VI, Qualification, is a live environment-training event under live-fire conditions using full-caliber training ammunition, realistic targets and engagement scenarios, at a combat tempo, conducted on an authorized live-fire facility.

Table 2-17. Example Crew Table VI

CREW TABLE V	Table I BASIC LIVE	Table II BASIC VIRTUAL	Table III INTERMEDIATE LIVE TADSS	Table IV INTERMEDIATE LIVE FIRE	Table V ADVANCED LIVE FIRE	Table VI ADVANCED LIVE FIRE
CREW PLATFORM	GST	STX-V	STX	BASIC	PRACTICE	QUAL
CREW QUALIFICATION-A commander's tool to assess the minimum proficiency of a crew during live fire that ensures the crews understands and demonstrates: • Their understanding of and compliance to the rules of weapons safety. • How to properly employ the weapons against most probable threats. • A general understanding of employment techniques of the primary and redundant optics and capabilities of the system. • The ability to defeat common threat presentations with the primary and redundant optics to a minimum standard. • A minimum level of performance that satisfies the Commander's requirements to assign the crew to the BATTLE ROSTER for readiness reporting.						

Table 2-17. Example Crew Table VI (continued)

CREW TABLE V	Table I BASIC LIVE	Table II BASIC VIRTUAL	Table III INTERMEDIATE LIVE TADSS	Table IV INTERMEDIATE LIVE FIRE	Table V ADVANCED LIVE FIRE	Table VI ADVANCED LIVE FIRE
CREW PLATFORM	GST	STX-V	STX	BASIC	PRACTICE	QUAL
 This event is required for readiness reporting by Army policy. Failure to follow the guidelines found in the training publication will compromise the qualification status of the crew. In general, the crew platform has a minimum of 2,400 basic engagement possibilities the crews should experience. The training strategy leverages simulations to provide those engagement experiences to each crew. Select, high-payoff engagements that demonstrate performance on critical engagement skills are conducted during crew live-fire training.						
GST-gunnery skills test or gunner's QUAL-qualification STX-situational training exercise STX-V-situational training exercise virtual TADSS-training aids, devices, simulators, and simulations						

Evaluation Procedures

2-61. Units are not authorized to present targets less than full-scale. All targets must meet the full-scale dimensions with standard thermal signatures as described in TC 25-8, *Training Ranges*.

2-62. Table VI is a performance-based evaluation of basic and advanced gunnery skills to the Army-standard of knowledge or skill in a live environment under live-fire conditions. Conduct of Table VI is on assigned vehicles using full-caliber ammunition, and battle effects simulators. It utilizes experienced evaluators to educate, train, test, and re-train service members to meet or exceed those standards.

2-63. The vehicle commander and gunner assigned as a crew have successfully demonstrated the skills and knowledge required to continue training on the platform through platform qualification. Battery commanders are confident in their service member's abilities in the position assigned. The tasks evaluated in Table VI provide the commander a means to certify the crew's basic and advanced weapons platform proficiency prior to collective live-fire events. The battalion commander and battalion unit trainer use the results from the qualification course when structuring the unit's crew training program and is a readiness reportable item.

2-64. Table VI will include 16 engagement scenarios- Two of which the Commander and Master evaluator can adjust to operational environments and constraints.

2-65. Units should plan for at least 3 training days per Battery allowing for range set-up maintenance and mechanical failure.

2-66. Master Gunners must be familiar with the family of maneuver ranges, the range operations control area (ROCA), and the specific instrumentation and target sets available on the range facility selected for Table VI. The maneuver range facilities recommended as primary and alternate facilities are shown below. See table 2-18 for Table VI task, conditions, and standards.

Table 2-18. Table VI, Task, Conditions, and Standards

Table VI, QUALIFICATION	
Task	Engage and destroy stationary and moving ground and aerial targets from a stationary, short halt (less than 30 seconds), or on-the-move, and degraded Man-portable air defense system (MANPADS) positions in a live fire environment.
Conditions	Given the following conditions a fully operational platform with weapon or weapon system, allotted ammunition, a certified Vehicle Crew Evaluator, and full-scale targets to meet the scenario requirements.
Standard	The crew must score a minimum of 700 out of 1000 points overall, score 70 points or more on all targets presented on seven out of ten engagements and qualify at least one-night engagement.

Table VI Qualification Engagements

2-67. The Master Evaluator/Commander has the authority to adjust engagement times as needed, with the goal of progressively increasing the level of difficulty with each training iteration and simulating realistic exposure times to various threats.

2-68. Targets must be fully exposed and locked out before engagement time starts. The commanders have choice on day/night/NBC engagements. See table 2-19 for M-LIDS crew proficiency requirements.

Table 2-19. Table VI, Mobile-low, slow, small unmanned aircraft integrated defeat system (M-LIDS) Crew Proficiency Requirements

ENG #	TARGET TYPE	TYPE OF ENGAGEMENT	TIME EXPOSURE	DAY/NIGHT	NUMBER OF ROUNDS
1	Troops in the open	240C COAX (GUNNER)	50 seconds	D/N	50
2	Aerial	Electromagnetic Attack	50 seconds	D/N	N/A
3	Aerial	.50Cal (GUNNER)	50 seconds	D/N	25
4	Aerial	COYOTE	50 seconds	D/N	1
5	Aerial	30MM (GUNNER)	50 seconds	D/N	20
	Troops in the open	240C COAX (GUNNER)	50 seconds	D/N	50
6	Aerial	Electromagnetic Attack	50 seconds	D/N	N/A
7	Moving Truck	.50Cal (GUNNER)	50 seconds	D/N	25
8	Pop Up Rotary	.50Cal (GUNNER)	50 seconds	D/N	25
9	Aerial	30MM (GUNNER)	50 seconds	D/N	20
	Troops in the open	240C COAX (GUNNER)	50 seconds	D/N	50
10	Aerial	Electromagnetic Attack	50 seconds	D/N	N/A
11	Aerial	30MM (GUNNER)	50 seconds	D/N	20
12	Aerial	30MM (GUNNER)	50 seconds	D/N	20
	Pop Up Rotary	.50Cal (GUNNER)	50 seconds	D/N	25
13	Aerial	Electromagnetic Attack	50 seconds	D/N	N/A
14	RPG Team	240C COAX (GUNNER)	50 seconds	D/N	50
	Light Truck FRT	.50Cal (GUNNER)	50 seconds	D/N	25
15	Aerial	COYOTE	50 seconds	D/N	1
16	Pop Up Rotary	.50Cal (GUNNER)	50 seconds	D/N	25
	Aerial	30MM (GUNNER)	50 seconds	D/N	20

Note. The engagement scenarios outlined in Tables II through VI involve electromagnetic attacks and Coyote interceptor engagements. However, if the range constraints do not permit the use of electromagnetic attack weapons alongside lethal defeat weapons, commanders may opt to skip the electromagnetic attack engagements for now. In such cases, they will adhere to the FS-LIDS guidelines to complete the electromagnetic attack gunnery tables at a later time.

Live firing Coyote interceptors is not permitted during training exercises. Instead, all Coyote engagements will be simulated using training aids, devices, simulators, and simulations also known as TADSS, provided that the system is available.

M-LIDS Advanced Gunnery Tables

2-69. These requirements apply to both practice and qualification tables. M-LIDS crew practice is a building block for crew qualification. Both exercises are fully resourced with ammunition. However, other training resource constraints may prohibit firing crew practice as prescribed.

2-70. Advanced gunnery tables incorporate at least eight collective tasks, the first five of which are mandatory.

2-71. The S3 identifies three additional collective tasks to support the unit METL, then develops an OPORD to support the mission.

2-72. The battalion unit trainer coordinates all the resources and personnel needed to support the gunnery exercise. Although this is a timed event, commanders may impose time standards to coincide with the unit METL.

2-73. Ground targets will be constructed IAW TC 25-8. Targets should also have signature devices simulate enemy fire. Targets should also have thermal signatures. Typical speed is in between 40 and 86 knots for fixed-wing aircraft. Three requirements support target engagements. The only acceptable aerial target is the one-fifth-scale target.

- For devise gunnery, one-fifth scale targets need PGS retroreflectors and smoke.
- For 30-mm live aerial engagements, targets need infrared (IR) source and miss-distance indicator (MDI) scoring systems.
- For each of the following weapons, place targets at the stated ranges:
 - 30-mm targets-500 to 1,600 meters
 - .50Cal targets-400 to 1400 meters
 - Coax targets-300 to 700 meters.
 - Coyote targets-1,000 to 5,000 meters.

Evaluation Standards

2-74. Commanders should integrate the tactical and gunnery tasks to exploit the full training effectiveness of these tables. Each crew of the platoon must earn at least 420 points on collective tasks. Each crew of the platoon must earn at least 280 points on gunnery tasks (aerial and ground targets combined).

- Vehicle exposure time in the defense must fall within 50 seconds.
- Each table uses the following 1,000-point system for gunnery.
 - Gunnery counts 1,000 points. The senior evaluator combines and averages the scores for aerial and ground targets. The senior evaluator uses the final numbers to determine a total score for that crew.
 - The senior evaluator combines and averages scores for both crews of the platoon to obtain a single, overall platoon rating.
- To qualify the crew's score must equal at least 700 points. The crew ratings are as follows.
 - Distinguished. Combined score of at least 900 points.
 - Superior. Combined score of 800 to 899 points.
 - Qualified. Combined score of 700 to 799 points.
 - Unqualified. Combined score of 699 or less, or failure to earn at least 420 points in gunnery.

Evaluation Team

2-75. The evaluation team consist of one officer who is a planner, another officer and one noncommissioned officer evaluates the overall tactical performance of a platoon.

2-76. The battalion S3 is responsible for the development, design, and execution of the exercise, with the assistance of the battalion master gunner. The battalion commander or his representative determines the overall training assessment of the crew.

2-77. One captain with battery-command experience should evaluate the platoon leader on overall collective task completion and mission success and conducts the platoon after-action-reviews.

2-78. A sergeant or staff sergeant with section sergeant experience should assist the evaluator. Each crew has one assistant, who observes overall crew performance, leader tasks, vehicle actions, and fire control and distribution: the sergeant or staff sergeant also helps with live-fire safety. The sergeant or staff sergeant assesses the crew during evaluation and provides feedback to crewmembers during the AAR.

2-79. The battalion commander may add friendly targets to advanced gunnery tables. The Battalion Commander may modify the types of tasks IAW unit METL. In the absence of a moving target, a stationary frontal target of the same type and at the same range is used. If a stationary target is used, target conditions must be adjusted.

Refires for Evaluation

2-80. A crew that fails to qualify must refire the failed engagements, but only until they earn enough points to qualify. The highest rating that a refiring crew can obtain is "Qualified". Battalion commanders decide whether to accept alibis.

2-81. The battalion commander should only accept alibis for target failures, weapon malfunctions, equipment failures not the result of crew error, and unsafe conditions not related to the firing vehicle.

FS-LIDS GUNNERY

2-82. FS-LIDS crews are defined as a group of personnel trained and certified to perform C-sUAS operation. FS-LIDS crew composition will depend on a range of factors, such as systems & personnel availability.

2-83. Table 2-20 shows a baseline of crew manning with individual responsibilities. Maintaining a crew smaller than the one outlined below is at the discretion of the Brigade Commander and requires a signed Memorandum to accept risk.

2-84. Crew members are encouraged to train/certify at multiple positions to increase redundancy. Non-battle rostered (i.e., low-density military occupational specialty) service members are encouraged to cross-train and familiarize themselves with various C-sUAS system when assets are available.

Table 2-20. FS-LIDS Crew Roles and Responsibilities

<i>Serial</i>	<i>Role</i>	<i>Rank</i>	<i>Description/Requirement</i>
1	Battle Captain, (IDA, TEA)	Officer	OIC of the crew. Responsible for all C-sUAS operations. The Battle Captain should hold written delegated identification authority (IDA) and target engagement authority (TEA) for all systems available for C-sUAS. Responsible for airspace deconfliction prior to engagement. Familiar with SPINs, ACO, ATO and other airspace control baseline documents. Understand and execute UAS self-defense criteria.
2	Battle NCO	NCO	Alternate Battle Captain including acting as IDA and TEA when the Battle Captain is unavailable. Responsible for the management of personnel on shift, ensuring maintenance of crew readiness. Knowledge of FAAD and C-UAS operations. Responsible to coordinate maintenance during system faults. Assist BLT CPT as required.
3	FAAD Operator	Any	Responsible for executing FAAD operations at the direction of the BTL CPT/ NCO, provides situational awareness of all tracks being surveilled by local sensors. Processes engagements at the discretion of the BTL CPT/NCO. Responsible for radio communications and maintaining the log of actions taken by the crew when not operating the FAAD.
4	Night Hawk/ Alternate FAAD Operator	Any	Night Hawk Camera and alternate FAAD Operator. Function in the same manner as the primary FAAD Operator. Provide undivided attention to Night Hawk feed or to the FAAD when on screen.
5	RTO	Any	Records radio transmissions and maintains the shift log (DA Form 1594, <i>Daily Staff Journal or Duty Officer's Log</i>). Relay messages from all parties on radio. Serve as runner for the C-sUAS crew
ACO-airspace control order ATO-air tasking order BTL-battle C-UAS-counter-unmanned aircraft systems FAAD-forward area air defense FS-LIDS-fixed site-low, slow, small unmanned aircraft integrated defeat system			IDA-identification authority SPINs-special instructions TEA- target engagement authority UAS-unmanned aircraft systems

FS-LIDS GUNNERY TABLES

2-85. This chapter discusses the basic and intermediate gunnery tables. All crewmembers will learn operator level maintenance, basic system knowledge, and operational procedures. Tables I through VI will be conducted by individuals and crews in preparation for individual certification followed by crew certification.

2-86. Certification validates that the units meet the required standards to perform their assigned wartime mission. Training is conducted per Department of the Army Pamphlet (DA PAM) 350-38 guideline standards, brigade-training strategy, and established crew drills as per this document. These training tables focus on preparing the individual to perform as part of a standardized crew. Table 2-21 outlines the gunnery tables for FS-LIDS.

Table 2-21. FS-LIDS Gunnery Tables

<i>Level</i>	<i>Table</i>	<i>Action</i>	<i>Frequency</i>	<i>Certifier</i>
Basic Gunnery (Individual)	I	FS-LIDS System/ Component Identification and Familiarization/PMCS/Maintenance/Crew Drills	Weekly	Unit Trainer
	II	VACR/FAAD C2/SKL/IFF Operations	Weekly	Unit Trainer
Intermediate Gunnery (Crew)	III	General Knowledge Test (SM must pass the test 30-days after being assigned to a crew)	Quarterly	Unit Trainer
	IV	Sense and Warm Validation/FS-LIDS System Link Validation	Monthly	Unit Trainer
Advanced Gunnery (Crew)	V	FS-LIDS Simulated Battle Drill Rehearsal	Monthly	BN Evaluator
	VI	FS-LIDS Simulated Battle Drills Qualifications	Quarterly	BN Evaluator
C2-command and control FAAD-forward area air defense FS-LIDS-fixed site-low, slow, small unmanned aircraft integrated defeat system IFF-Identification, friend or foe PMCS-preventive maintenance checks and services SKL-simple key loader VACR-visual aircraft recognition				

Certification Authority and Frequency

2-87. Below is the certification authority and frequency of each evaluation depending on the gunnery table level. The frequency of the training provided is the minimum baseline requirement. Units are encouraged to train above the baseline frequency.

2-88. Tables I-II (basic gunnery) and Table III-IV (intermediate gunnery) are evaluated on individuals by unit level trainer. Tables I-II should be trained weekly, and Table III tested quarterly. The unit trainer is the certification authority. All FS-LIDS crew members must pass the C-sUAS General Knowledge Test within 30-days of being assigned to a FS-LIDS crew and quarterly thereafter. A score of 80-points or above is consider pass and unit trainer will keep hard copies of the test as historical training record. Service members that score below 80-points will conduct retraining given by the unit trainer immediately after their failure and re-test within the next 24-hrs.

2-89. Tables V-VI (Advanced Gunnery) are evaluated on crews by battalion evaluators with battalion commander being the certification authority. Table V should be trained monthly with Table VI certified quarterly. After the completion of Tables I-V, the unit commander will sign a memorandum for record with attached battle roster stating the composition, dates of training, and scores of each crew. Once signed, the memorandum enables the crews to request Table VI qualification by the battalion evaluator.

Table I

2-90. This table is used to train operators on the fundamentals of the FS-LIDS system operation. Table I consist of FS-LIDS components identification, capabilities/limitations, proficiency in performing preventative maintenance checks and services (PMCS), pre-engagement drills and post-engagement drills on

their assigned systems. All operators must be licensed on supporting power generation equipment where applicable.

2-91. Operator level maintenance should be conducted on all subsystems to include generators and computer systems. All operators are required to understand the mission impact when an FS-LIDS component become NMC so they can report accurately to higher echelon.

2-92. Low-density military occupational specialty service members are strongly recommended to attend FS-LIDS Table I training to ensure their situational awareness of the system sustainment requirements. See Table 2-22 for the task, conditions, and standards for Table I.

Table 2-22, Table I, Task, Conditions, and Standards

TASK:	Identify Systems and Subsystem components, capabilities, and limitations. Conduct preventive maintenance checks and services (PMCS) of all systems that comprise the fixed site-low, slow, small unmanned aircraft integrated defeat system (FS-LIDS) system architecture and crew drills. Understand system maintenance/sustainment requirements.
CONDITIONS:	Individual service members will identify and discuss systems given all equipment, technical manuals (TM), standard operating procedures, baseline special instructions, and rules of engagement. Service members will also conduct PMCS and have knowledge of PMCS standards according to TMs.
STANDARDS:	Each service member must be able to identify the systems on which they are assigned as well as the subsystem components that comprise each system. They must also be able to describe in detail the capabilities that each subsystem provides to the counter-small unmanned aircraft systems (C-sUAS) system architecture as well as the limitations of those systems. Maintainers and Operators are responsible for conducting PMCS IAW Tactical Manual (TM) or Non-program of record User Manual with no deficiencies and conduct individual PMCS training with no deficiencies. Crews must conduct pre-engagement drills and post-engagement drills in their assigned role with no deficiencies or safety violations. A NO-GO in any area of Table I constitutes Table I failure and operator must be re-tested.

Table II

2-93. Table II evaluates the operator's proficiency in visual aircraft recognition, forward area air defense (known as FAAD) command and control (C2), electro-optical cameras, IR cameras, and identification of friend or foe (IFF).

2-94. The crew members train to recognize enemy and friendly aircraft. Crew members also receive instruction on FAAD C2 / EO/IR cameras / SKL operations, charging the IFF programmer batteries, and IFF code loading procedures. See Table 2-23 for the task, conditions, and standards for Table II.

Table 2-23. Table II, Task, Conditions, and Standards

TASK:	Demonstrate technical proficiency in visual aircraft recognition (VACR) using assigned counter-small unmanned aircraft systems (C-sUAS) equipment.
CONDITIONS:	Individual service members will identify and operate systems given all equipment, TM's, SOP's, baseline special instructions, and rules of engagement.
STANDARDS:	All crew members must pass a written VACR test, and each crew member must demonstrate proficiency in manipulating the FAAD C2 and electric optical/infrared (EO/IR) cameras, successfully program the IFF to accept a code, and/or demonstrate proficiency in other system procedures and battery charging depending on their assigned role.
C2-command and control FAAD -forward area air defense IFF-Identification, friend or foe SOP-standard operating procedure TM-technical manual	

Table III

2-95. This table is a critical gate used to provide an evaluation for the FS-LIDS crew members on Tables I-II. The C-sUAS general knowledge test is used to evaluate crewmember's proficiency for Table I-II.

2-96. All FS-LIDS crew members must pass the general knowledge test within 30-days of being assigned to a FS-LIDS crew. A score of 80-points or above is consider passing and unit trainer will keep hard copies of the test as historical training record.

2-97. Service members that scored below 80-points will conduct retrain by the unit trainer immediately after their failure and re-test within the next 24-hrs. See Table 2-24 for the task, conditions, and standards for Table III.

Table 2-24. Table III, Task, Conditions, and Standards

TASK:	Pass an evaluation of Tables I and II within a 30-day period.
CONDITIONS:	Individual service members will identify, operate, conduct maintenance, and perform crew drills on C-sUAS systems given all equipment, TM's, SOP's, AADP, baseline SPINS, and ROE.
STANDARDS:	Pass an evaluation of C-sUAS general knowledge test with a score greater than or equal to 80%, pass Table II with a score greater than or equal to 80% within a 30-day period.
C-sUAS-counter-small unmanned aircraft systems ROE-rules of engagement SOP-standard operating procedure SPINS-special instructions	

Table IV

2-98. This table is a validation of indoor and outdoor alarm systems, unit response time to the test of the alarms. It also tests the crews' ability to integrate FS-LIDS into the FAAD network and maintain green communication status across all systems. NIPR/SIPR VOIP connections are tested with unit's higher headquarters and other adjacent units.

2-99. FAAD operators should be able to select the correct alarm for a given scenario, RTOs should be able to effectively operate the push-to-talk microphone, and units should ensure that alarms can be heard in all locations in which service members would be located.

2-100. Crews need to be able to understand initialization processes and isolating faults in their local network architecture to include validating physical connections.

2-101. FS-LIDS crews validate NIPR/SIPR account to ensure they can send reports on proper channels. Table VI is conducted monthly to maintain communication readiness. See Table 2-25 for the task, conditions, and standards for Table IV.

Table 2-25. Table IV, Task, Conditions, and Standards

TASK:	Conduct validation of Sense and Warn systems, unit response to sense and warn, and link/communication validation.
CONDITIONS:	All systems are emplaced and integrated into the FAAD C2. React to Indirect Fire (IDF) and React to Enemy Aircraft Drills have been codified and previously rehearsed. All C-sUAS systems connected and maintain green status on FAAD. C-sUAS FAAD C2 tied into JDN network. Operators have working NIPR/SIPR accounts. NIPR and SIPR phone connected with higher echelon units and air space control authority.
STANDARDS:	System Operators successfully operate indoor alarms, outdoor alarms, and loudspeaker systems. Personnel react to the alarms and report up accountability IAW unit SOP. All FS-LIDS systems maintain green status on FAAD. BDOC FAAD received JDN Tracks. Primary/Alternate Voice & Data contacted and maintained.
BDOC-base defense operations center C2- command and control C-sUAS-counter-small unmanned aircraft systems FAAD-forward area air defense FS-LIDS-fixed site-low, slow, small unmanned aircraft integrated defeat system JDN-Joint Data Network SOP-standard operating procedure	

Table V

2-102. Table IV trains the FS-LIDS crews to execute simulated battle drills in a training environment. Scenarios should be reflective of both the most likely course of action and most deadly course of action.

2-103. Table IV is an internal evaluation given by the unit trainer 10 out of 12 battle drills must be executed successfully to receive a GO. Information on designing scenarios can be found later in this chapter. See table 2-26 for the task, conditions, and standards for Table IV.

Table 2-26. Table IV, Task, Conditions, and Standards

TASK:	Fixed site-low, slow, small unmanned aircraft integrated defeat system (FS-LIDS) crew executes battle drill rehearsals.
CONDITIONS:	FS-LIDS systems are fully emplaced with network connectivity and receive simulated scenarios, while being evaluated by internal source.
STANDARDS:	FS-LIDS crew successfully executes 10 out of 12 simulated battle drills.

Table VI

2-104. Table VI evaluates the FS-LIDS crews to execute simulated battle drills in a training environment. Scenarios should be reflective of both the most likely course of action and most deadly course of action.

2-105. Table VI is an external evaluation given by the battalion evaluator. 10 out of 12 battle drills must be executed successfully to receive a GO. Information on designing scenarios can be found later in this chapter. See Table 2-27 for the task, conditions, and standards for Table VI.

Table 2-27. Table VI, Task, Conditions, and Standards

TASK:	Fixed site-low, slow, small unmanned aircraft integrated defeat system (FS-LIDS) crew executes simulated battle drills.
CONDITIONS:	FS-LIDS systems are fully emplaced with network connectivity and receive simulated scenarios, while being evaluated by external source.
STANDARDS:	FS-LIDS crew successful execute 10 out of 12 simulated battle drills.

FS-LIDS BATTLE DRILLS

2-106. The FS-LIDS battle drills test crews' ability to conduct C-sUAS operations while requiring crewmember understand basic air de-confliction. The battle drills are built on realistic scenarios and historical data. Unit trainers are required to refine battle drills and develop new battle drill for C-sUAS training when the enemy evolve with new UAS tactics, techniques and procedure.

2-107. Battle drills are broken down into three categories.

- C-sUAS engagements (Battle Drill 1-5).
- Friendly protect and system faults (Battle Drill 6-8, 11 and 12)
- Complex attacks (Battle Drill 9-10)

2-108. Units should use the crawl, walk, and run approach to ensure all necessary skills are developed as crews progress through each battle drill. All battle drills will be tested prior to Table VI to ensure all requirements are met within the scenario.

Battle Drill Guidelines

2-109. The following guidelines are to be used when creating battle drill scenarios. Units are permitted to increase the number of targets or variables contained in the battle drill tables to increase the difficulty of their training.

2-110. When creating scenarios, units should incorporate their intelligence preparation of the operational environment to ensure systems and crews are capable of protecting defended assets against the enemy's most likely course of action and most dangerous course of action.

2-111. On systems classified SECRET, the FAAD should be configured to match the FOB size, quantity and locations of FS-LIDS defended assets, primary target line of lethal defeat systems, and data link architecture. Digital Terrain Elevation Data level-2 should be loaded into simulator systems to provide realistic training possible to C-sUAS crews. Scenarios should have accurately modeled aircraft to aid in visual aircraft recognition training.

2-112. On UNCLASSIFIED systems, the FAAD should be configured in a generalized way that allows crews to train with all combinations of sensors and effectors against threat representative aircraft. Units will avoid inputting data which contains force compositions, locations, or any other classified information onto UNCLASSIFIED systems. Scenarios should have accurately modeled aircraft to aid in visual aircraft recognition training.

2-113. Scenarios will be made up of friendly, neutral, and suspect aircraft and rockets, artillery, and mortars. The number of air tracks, UAS and RAM are listed on table 2-28 on page 2-26. Units should be in continual contact with the intelligence community and adjacent units to ensure target sets are representative of past threats, present threats, and possible future threats.

2-114. Scenarios will contain a variety of aircraft within the FAAD sub-boundary IAW normal daily activity within the simulated area. When applicable, scenarios should accompany a simulated air tasking order and "squawk" the appropriate modes and codes. Where applicable, aircraft should populate their classifications over external connection based on track parameters, track correlation, and manual classification by external authority.

2-115. Battle drill scenarios are used for air battle management training by all units. Each unit will maintain a minimum of one complete set of battle drills for training. Units are highly encouraged to work together when creating battle drill libraries. The battalion evaluator will periodically review battle drill scenarios to ensure training quality and intensity across all subordinate units. Battalion evaluators must be trained on the systems utilized for training to assist with creating realistic scenarios.

Table 2-28. Fixed Site-Low, Slow, Small Unmanned Aircraft Integrated Defeat System (FS-LIDS) Battle Drills

Battle Drill #	# of UAS/RAM	Description
1	Unmanned aircraft system (UAS): 1x Group 1 Quadcopter RAM: None	Single Quadcopter loitering, hostile intent shown. Start video recording; engage w/EW.
2	UAS: 2x Group 1 Quadcopter RAM: None	Single Quadcopter loitering, hostile intent shown. Start video recording; engage w/EW. UAS resistant to EW, engage w/KD.
3	UAS: 1x Group 2-3 RAM: None	Single one-way attack UAS approaching. de-conflict airspace and engage w/EW & KD
4	UAS: 3-4 x Group 2-3 RAM: None	Multiple one-way attack UAS approaching, de-conflict airspace and engage w/EW & KD
5	UAS: 2x Group 2-3 1x Friendly RW	Multiple one-way attack UAS approaching. Clear Airspace. Once Airspace CLEAR, engage w/KD
6	UAS: 1x Neutral Group 2-3	Neutral UAS inside or outside the Base Defense Zone (BDZ). Record and monitor Neutral Track
7	UAS: 1x UAS 1x Friendly RW	Friendly manned or unmanned aircraft inside the BDZ while it has been marked "Hot" due to ongoing engagement.
8	UAS: 1x UAS / 1 or 2 Equipment	System fault occurs prior to or during C-sUAS engagement. Fault can be any combination of radars, IFF capabilities, EO/IR cameras, EW effectors, launchers, or missiles
9	UAS: 1x OWA-UAS RAM: 2x IDF track	Combined use of ENY UAS and IDF. UAS may be conducting ISR for the IDF or direct attack to overwhelm base. React to IDF & UAS w/KD
10	UAS: 2-3 OWA-UAS RAM: 2x IDF track on fob, 2x Rocket IDF over FOB	Multiple one-way attack UAS approaching. Engage w/KD. Issue bunker order and react to IDF Sound S&W for IP-Out Track
11	1x Unknown UAS turns Friendly	Unknown UAS track lost link during ID process; BDOC stand by for regaining link and ID track correctly
12	1x Clutter Track	Unknown track pop-up near BDZ. VACR the track to be biological or balloons BDOC remove protection state from clutter track

Note. The engagement scenarios outlined in Tables II through VI involve electromagnetic attacks and Coyote interceptor engagements. However, if the range constraints do not permit the use of electromagnetic attack weapons alongside lethal defeat weapons, commanders may opt to skip the electromagnetic attack engagements for now. In such cases, they will adhere to the FS-LIDS guidelines to complete the electromagnetic attack gunnery tables at a later time.

Live firing Coyote interceptors is not permitted during training exercises. Instead, all Coyote engagements will be simulated using training aids, devices, simulators, and simulations also known as TADSS, provided that the system is available.

BDOC-base defense operations center

BDZ-base defense zone

EW-electronic warfare

FS-LIDS-fixed site-low, slow, small unmanned aircraft integrated defeat system

IDF-indirect fire

IFF-Identification, friend or foe

UAS- unmanned aircraft system

VACR-visual aircraft recognition

Chapter 3

Fire Control Process

Chapter 3 covers the procedures used from the time targets are acquired, through the placement of fires on those targets, to the reporting of the effects of those fires to the unit commander. Also included are considerations for fire distribution and control during offensive and defensive operations. Although the discussion focuses on the actions at the platoon and section level, these actions are always integrated into and become part of the unit plan.

EFFECTIVE FIRE CONTROL

3-1. Effective fire control requires a unit to rapidly acquire the enemy and mass the effects of fires to achieve decisive results in the close fight. When planning and executing direct fires, the commander and subordinate leaders must know how to apply several fundamental principles. The purpose of these principles of direct fire is not to restrict the actions of subordinates. Applied correctly, these principles help the unit accomplish its primary goal in any direct fire engagement—to both acquire first and shoot first giving subordinates the freedom to act quickly upon acquisition of the enemy. This discussion focuses on the following principles:

- Mass the effects of fire.
- Destroy the greatest threat first.
- Avoid over engagement of a target.
- Employ the best weapon for the target.
- Minimize friendly exposure.
- Prevent fratricide.
- Plan for limited visibility conditions.
- Develop contingencies for diminished capabilities.

MASS THE EFFECTS OF FIRE

3-2. Units must mass its fires to achieve decisive results. Massing entails focusing fires at critical points and distributing the effects. Random application of fires is unlikely to have a decisive effect. For example, concentrating the unit's fires at a single target may ensure its destruction or suppression; however, that fire control technique will probably not achieve a decisive effect on the enemy formation or position.

ASSESS AND DESTROY THE MOST DANGEROUS THREAT FIRST

3-3. The order in which the unit engages enemy forces is in direct relation to the danger they present. The threat posed by the enemy depends on the weapons, range, and positioning. Presented with multiple targets, a unit will, in almost all situations, initially concentrate fires to destroy the greatest threat, and then distribute fires over the remainder of the enemy force.

AVOID OVER ENGAGEMENT OF A TARGET

3-4. Use only the amount of fire required to achieve necessary effects. Over engagement of targets waste ammunition and saturates weapons that are better employed acquiring and engaging other targets. The idea of having all weapon systems engage a different target must be tempered by the requirement to destroy the greatest threats first.

EMPLOY THE BEST WEAPON FOR THE TARGET

3-5. Using the appropriate weapon for the target increases the probability of rapid enemy destruction or suppression and saves ammunition. The unit has many weapons to engage the enemy with. Target type, range, exposure, weapons and ammunition availability, and desired target effects are key factors in determining the weapon or electromagnetic warfare effector that should be employed. Additionally, leaders should consider individual crew capabilities when deciding on the employment of weapons. The commander will task organize and array their forces based on the terrain, enemy, and desired effects of fires.

MINIMIZE FRIENDLY EXPOSURE

3-6. Units increase their survivability by exposing themselves to the enemy only to the extent necessary to engage them effectively. Natural or man-made defilade provides the best cover from direct fire munitions. Crews and squads minimize their exposure by constantly seeking effective available cover, attempting to engage the enemy from the flank, remaining dispersed, firing from multiple positions, and limiting engagement times.

PREVENT FRATRICIDE

3-7. All personnel must be proactive in reducing the risk of fratricide and noncombatant casualties. There are numerous tools to assist in this effort. They are identification training for combat vehicles and aircraft, the unit's weapons safety posture, the weapons control status, recognition markings, and a common operational picture. Knowledge and employment of applicable rules of engagement are the primary means of preventing noncombatant casualties.

DEVELOP CONTINGENCIES FOR DIMINISHED CAPABILITIES

3-8. Leaders initially develop plans based on their unit's maximum capabilities; they make alternate plans for implementation in the event of casualties or weapon damage or failure. While leaders cannot anticipate or plan for every situation, they should develop plans for what they view as the most probable occurrences. Building redundancy into these plans, such as having two systems observe the same sector, is an invaluable asset when the situation (and the number of available systems) permits. Designating alternate sectors of fire provides a means of shifting fires if adjacent elements are knocked out of action. Examples of contingencies for diminished capabilities include-

- Designating certain C-sUAS systems to engage certain targets due to ammunition shortages or weapon malfunctions.
- Designating certain C-sUAS systems within the combined arms platoon to engage certain targets due to inherent weapon range or firepower limitations.

TARGET REFERENCE POINT

3-9. A target reference point (TRP) is a recognizable point on the ground that leaders use to orient friendly forces and focus and control direct fires. In addition, when TRPs are designated as indirect fire targets, they can be used in calling for and adjusting indirect fires. Leaders designate TRPs at probable enemy locations and along likely avenues of approach. These can be natural or man-made points. A TRP can be an established site, such as a hill or a building, or an impromptu feature designated as a TRP on the spot, like a burning enemy vehicle or smoke generated by an artillery round. Friendly units can also construct markers to serve as TRPs. Ideally, TRPs should be visible in three observation modes (unaided, passive-IR, and thermal) so they can be seen by all forces. Example of TRPs include the following features and objects:

- Prominent hill mass.
- Distinctive building.
- Observable enemy position.
- Destroyed vehicle.
- Ground-burst illumination.
- Smoke round.

ENGAGEMENT AREA

3-10. This fire control measure is an area along an enemy avenue of approach where the commander intends to mass the fires of available weapons to destroy an enemy force. The size and shape of the engagement area is determined by the degree of relatively unobstructed inter-visibility available to the unit's weapon systems in their firing positions and by the maximum range of those weapons. Typically, commanders delineate responsibility within the engagement area by assigning each platoon a sector of fire or direction of fire; these fire control measures are covered in the following paragraphs.

SECTOR OF FIRE

3-11. A sector of fire is a defined area that must be covered by direct fire. Leaders assign sectors of fire to subordinate elements, crew-served weapons, and individual service members to ensure coverage of an area of responsibility. Leaders may also limit the sector of fire of an element or weapon to prevent accidental engagement of an adjacent unit. In assigning sectors of fire, commanders and subordinate leaders consider the number and types of weapons available. In addition, they must consider acquisition system type and field of view in determining the width of a sector of fire. Conversely, most fire control acquisition systems have greater detection and identification ranges than the unaided eye, but their field of view is narrow. Means of designating sectors of fire include-

- TRPs.
- Clock direction.
- Terrain-based quadrants.
- Friendly-based quadrants.

DIRECTION OF FIRE

3-12. A direction of fire is an orientation or point used to assign responsibility for a particular area on the battlefield that must be covered by direct fire. Leaders designate directions of fire for the purpose of acquisition or engagement by subordinate elements, crew-served weapons, or individual service members. Direction of fire is most employed when assigning sectors of fire would be difficult or impossible because of limited time or insufficient reference points. Means of designating a direction of fire include-

- Closest target reference point.
- Clock direction.
- Cardinal direction.
- Tracer on target.
- Infrared laser pointer.

QUADRANTS

3-13. Quadrants are subdivisions of an area created by superimposing an imaginary pair of perpendicular axes over the terrain to create four separate areas or sectors. Quadrants can be based on the terrain, friendly forces, or enemy formation.

3-14. The method of quadrant numbering is established in the unit SOP; however, care must be taken to avoid confusion when quadrants based on terrain, friendly forces, and the enemy formations are used simultaneously.

TERRAIN-BASED QUADRANT

3-15. A terrain-based quadrant entails use of a target reference point, either existing or constructed, to designate the center point of the axes that divide the area into four quadrants. This technique can be employed in both offensive and defensive operations. In the offense, the commander designates the center of the quadrant using an existing feature or by creating a reference point. The axes delineating the quadrants run parallel and perpendicular to the direction of movement. In the defense, the commander designates the center of the quadrant using an existing or constructed target reference point.

3-16. Quadrants are marked using the letter "Q" and a number (Q1 to Q4); quadrant numbers are in the same relative positions as on military map sheets (from Q1, as the upper left-hand quadrant, clockwise to Q4, as the lower left-hand quadrant).

This page intentionally left blank.

Appendix A

Planning and Establishing Ranges for C-sUAS Gunnery

This chapter outlines procedures, duties, and responsibilities for planning and establishing gunnery ranges and tactical courses, to include developing scenarios for gunnery ranges to support C-sUAS. A combat training program is conducted on training areas to include dry and live firing. It provides an opportunity to acquire targets in a realistic environment and use weapon systems to engage targets. Tactical training should be integrated with gunnery training.

PLANNING GUNNERY EXERCISES

A-1. When planning a C-sUAS gunnery exercise the following procedures are followed:

- Determine the commander's intent for the end state of the exercise.
- Make tentative plans for crews and collective gunnery.
- Evaluate range and ammunition resources to support the plans.
- Forecast resources necessary to execute the training event(s).
- Develop scenarios for crew and collective gunnery tables.
- Proof all scenarios prior to execution.
- Coordinate with supporting elements.
- Plan range operations and personnel.
- Execute the exercise.
- Evaluate the exercise through after-action reviews.

COMMANDER'S INTENT

A-2. Planning a gunnery training exercise begins with the commander's intent for the desired end state of the exercise. The commander will issue guidance to their staff on the training mission(s) that the unit is to perform. The staff first plans for the collective gunnery phase to meet the commander's intent. The staff tailors a gunnery training program that supports the collective training plan. The S-3 and unit trainer develop the gunnery program using their inherent flexibility to tailor the tasks to the mission to meet the commander's intent.

A-3. To develop the gunnery program:

- The S-3 and unit trainer identify how crew tables and tasks are going to be designed to train to the mission that will be used in the collective gunnery evaluations.
- Unit trainers determine which ranges will support firing these tables, tasks and develops scenarios for each table.
- The commander and S-3 determine which core mission(s) to conduct based on the unit's METL and specific command guidance.
- The S-3 identifies the primary collective tasks to evaluate based on the core missions.
- The S-3 identifies which collective tasks will be firing tasks.
- The S-3 and unit trainer develop scenarios that require sections or platoons to conduct designated core missions and respective primary collective tasks.

SITE SELECTION

A-4. Before selecting a site, identify the training area or range(s) that will support the C-sUAS training event. Make a thorough map and ground reconnaissance of areas available for firing and electronic attack. The range must be large enough to accommodate all weapon systems, types of ammunition, radio frequencies to be used and jammed, and types of weapons to be fired. Some considerations for site selection are:

- Sufficient maneuver area and enough targets to provide several routes and target arrays.
- Targets in realistic arrays and, where possible, not marked by berms.
- Enfilade and defilade firing positions.
- Sand table layouts of the entire range area to aid rehearsals.

- What equipment might be affected by electromagnetic attack methods.
- Coordination with range operations to ensure select range will meet all requirements.
- Terrain that accommodates integration of squad and vehicle fire and maneuver.
- All jamming systems require electromagnetic attack (EA) request to train and operate.
- Missile projectiles are approved for flight only at major range facility test bases.
- Spectrum approval required to operate on network for electromagnetic warfare effectors.

EA Clearance Request Coordination Requirements

A-5. Army commands are overall responsible for submission of each command EA request in official command memorandum to the Army Spectrum Management Office. Before submitting a command EA request, the originator (drafter) of the EA request (memorandum) shall identify in detail the testing unit's EA requirements as stated in CJCSM 3212.02E *Performing Electromagnetic Attack in the U.S. & Canada for Tests, Training, and Exercises*.

A-6. Coordination requirements for EA testing and training in the United States is based on the desired EA frequency band, geographical area, proposed duration, time of the EA operation, and EA jamming equipment specifications.

A-7. Detailed information assists national coordination agencies in assessing the impact of EA testing and training operations on the RF spectrum and expedites the approval process.

A-8. The EA clearance approval process takes approximately 60 days to complete.

Range or Tactical Course Reconnaissance

A-9. The S-3 and unit trainer conduct a reconnaissance of the ranges they will use to support their gunnery training plan. Additionally, the officer in charge and noncommissioned officer in charge for each range should personally conduct a reconnaissance and coordinate with range control headquarters before the unit occupies the range or training area. The reconnaissance should provide answers to the following questions:

- What route to the range or training area will be used?
- Are enfilade and defilade positions available?
- What control facilities are available? What is their condition?
- Is the tower equipped with sufficient communication equipment?
- Are range safety markers visible for live fire? How will they be illuminated at night?
- How is access to the impact area controlled for live fire? What are guard requirements?
- Does the range support firing of all types of ammunition and pyrotechnics require for the exercise?
- Who furnishes targets, target supplies.
- What is the condition of target mechanisms?
- Who provides the target operators and target details?
- What is the condition of moving targets?
- Has the range or training area been cleared of unexploded ordnance?
- Who will furnish firefighting equipment, range flags, and range regulations?
- Does the range or training area provide adequate space for maneuvering vehicles and the weapons to be used?

A-10. Here are the following administrative areas:

- Ammunition pad?
- Clearing pit and misfire pit?
- Barriers/guard posts?
- Mess area?
- Latrine?
- Helipad?
- Aid station?
- Troop break and billet areas?
- Briefing and debriefing tent (after action review facility)?
- Concurrent training area?

Range or Training Area Layout

A-11. Coordinate with the range facility manager to gather information about the facility. Determine if the surface danger zone diagram is current, if not construct one if necessary. As a minimum, you will need to obtain the following information about your facility:

- A scaled range diagram (preferably in 100-meter increments) that identifies target pits (by type and number), battle positions, and firing points or map of the training area.
- A fire/no-fire matrix (if available), which is the authorized list of targets allowed to be fired within the surface danger zone diagram from each firing position, typically located in the range SOP.
- Conduct a reconnaissance of the facility to:
 - Identify any restrictions for the facility.
 - Determine the best lanes that allow the most targets to be engaged.
 - Determine if the facility supports dual-lane scenario or only single-lane firing.
 - Plan to maximize throughput.

A-12. Answer the following questions before developing the scenario:

- How many lanes will be firing?
- What weapon platform types will be firing or emitting EA?
- Is there adequate airspace for C-sUAS targets to safely maneuver and recover?
- What is the maximum distance of the firing box for offensive engagements?
- What is the order of tasks for each scenario?
- Is cross-firing allowed on the range?

Note. The order of tasks is usually most influenced by the location of specific target types on the range, typically moving targets or targets in urban clusters. Plan these engagements first when developing a scenario.

DEVELOPING THE SCENARIO

A-13. Once planning is complete, develop the scenario, including the following:

- Identify the first firing position and the first task to be fired.
- When listing target information, include the following:
 - Target pit number or target number.
 - Target type.
 - Range.
 - Exposure time (target lift and delay time, if applicable).
 - Hostile fire signature (no less than five seconds after exposure).
 - Type of ammunition to be fired at the target.
 - Number of hits required to kill the target.
 - Target speed (if moving).
 - Target direction (if moving).
 - Evasive moving target (plan maneuver).
 - Alternate target number (if applicable).

TARGETRY

A-14. All C-sUAS targets are constructed according to TC 25-8. Moving targets should be presented at speeds that emulate known threats.

A-15. Currently, an evasive target can be programmed to move at varying speeds, conduct short halts, or change direction. They must be visible from the firing position for the entire exposure time and must run for a minimum of 15 seconds in each step to allow for acquisition, tracking, and engaging.

A-16. Evasive targets should have the ability to change speed and direction and alter their appearance or thermal signature as the vehicles' directions change. All target types must have thermal target signatures for both day and night engagements according to TC 25-8. The type of targets that will be used are aerial targets. Aerial targets will be presented as moving, frontal, or flank threat attacks.

A-17. C-sUAS targets must have an air worthiness release as prescribed in AR 70-62 *Airworthiness of Aircraft Systems*. This regulation prescribes policies, responsibilities, processes, and procedures for life cycle airworthiness (design, production, and continued) of manned and unmanned aircraft systems and subsystems, including the installation of allied equipment and modifications to Army aircraft. This regulation implements DoDI 5030.61 *DoD Airworthiness Policy*. Army aircraft as used herein includes all aviation materiel and aircraft that is Army assigned, bailed, borrowed, loaned, leased, owned, or otherwise authorized for operation by Army personnel, or after modification under an Army contract.

A-18. C-sUAS targets must have explicit authority granted within their air worthiness release to:

- Fly over personnel.
- Fly over vehicles.
- Be flown out of line of sight.
- Be flown in the rain.

FRIENDLY AND NEUTRAL PRESENTATIONS

A-19. Realistic friendly and neutral presentations may be incorporated in all phases of gunnery to train crewmembers in combat identification and fratricide avoidance. Presentations must appear realistic in visual, thermal, and IR optics. Friendly or neutral presentations will be marked as follows:

- Friendly vehicle. Friendly vehicle presentations will be displayed with combat identification panel (known as CIP) properly located on the silhouette. Phoenix beacons should be used at night to provide IR optics with friendly identification signature. Unit SOPs can also be used to develop specific identification markings.
- Friendly troop. Units should use established SOPs to depict friendly troop thermal signature combat identification for night engagements. Phoenix beacons should be used at night to provide IR optics with friendly identification signature.
- Friendly UAS. Friendly UAS may be equipped with IR markings or emitters that can be detected by IR sensors. They may also display visual markings, such as colors, symbols, or patterns, to identify them as non-hostile. These markings can be applied to the drone's body or wings.
- Neutral. Neutral presentations must be clearly identifiable as civilian vehicles and personnel. Personnel presentations may consist of any number silhouettes dispersed in a small group.

SURFACE DANGER ZONE DIAGRAMS

A-20. When establishing ranges, units must submit surface danger zone diagrams to the installation range control facility for approval before firing. Restrictions and precautions for surface danger zone diagrams are found in DA Pam 385-63 (*Range Safety*) and TB Med 524 (*Occupational and Environmental Health: Control of Hazards to Health from Laser Radiation*). Surface danger zone diagrams show range boundaries and safety features in overlay form, including safety limit markers for each firing position. Firing tables provide values for range, maximum ordinates, electromagnetic spectrum management, and RF hazards.

Note. If you must construct a surface danger zone diagram, refer to DA Pam 385-63.

PROOFING THE SCENARIO

A-21. After the C-sUAS scenario is approved on paper, it must be proofed on the facility using the system(s) for which the scenario was approved. It is imperative that this be conducted before the unit arrives, to prevent lost training time. All conditions must be the same as if firing, using the same type of weapon platform(s) that will be firing.

A-22. If firing a dual-lane scenario, proof both lanes at the same time. Each target should be checked for validity by making sure it can be viewed through the firing systems optics throughout the entire presentation time. The range to targets and the target presentation times should be verified.

Developing Scenarios for Collective Gunnery Tables

A-23. CATS, along with scenario development, the evaluation criteria, and requirements in DA PAM 350-38 Standard in Training Commission, establish an Army standard for training and evaluation. At the same time, they allow unit commanders to tailor tactical tasks based on their missions and training emphasis.

A-24. These tables should be designed to be combined arms events, to employ multi-echelon training, and to be interactive to the extent that tactical decisions affect targetry presentations. This section provides information for developing tactical scenarios to support collective gunnery training.

A-25. Collective proficiency exercises and practice tables are a building block for section and platoon qualification. Sections and platoons should collectively participate in device-based training at least once before conducting live-fire training at that level.

A-26. When they do qualify, sections and platoons follow the guidelines and standards in the applicable CATS and this appendix. The commander and their S-3 determine which CATS core mission(s) to conduct based on the unit's METL and specific command guidance.

A-27. The S-3 identifies the primary collective tasks to evaluate based on the core missions. The S-3 identifies which collective tasks will be firing tasks. Section and platoon gunnery will be conducted in two phases, one during the day and one at night.

A-28. The S-3 and master gunners develop scenarios that require the sections and platoons to conduct designated core missions and their respective primary collective tasks. Live- fire gunnery presentations are incorporated into the firing tasks. This appendix identifies the type and the minimum number of live-fire targets for each platform system during qualification.

A-29. However, units may add targets if resources allow. The S-3, S-2, and master gunners select troop arrays that represent threat forces in the operating environment or the required training emphasis. They also specify threat target types and engagement distances, or range bands as well as most-to-least-dangerous targets, based on the threat analysis and training emphasis.

A-30. The end product is a threat template for section and platoon gunnery exercises. The S-3 develops an OPOD that supports the missions. The master gunner prepares evaluation packets for each section or platoon. Each packet contains training and evaluation outlines, as well as tactical and gunnery score sheets. The master gunner arranges the contents in collective-task sequence and prepares an all-inclusive list of what is in an evaluation packet.

A-31. The master gunner coordinates all necessary resources and personnel to support the gunnery exercises.

A-32. Graphics used to control movement to the range and section sectors of fire and axis of advance (range limitations) will be developed as a hard copy and transmitted, if equipped, via digital communications. Section and platoon leaders must develop their own additional direct fire control and distribution measures and graphics using the platoon SOP. Situation reports and fragmentary orders used to control the section will be issued based on these graphics.

Note. Collective gunnery training can be enhanced by integrating combined arms training to the gunnery exercises. Each participating unit, with its specific weapon systems, must be included in the planning process.

A-33. For range planning and development requirements including required personal, range control briefing, and administration and emergency direction, refer to DA PAM 385-63.

This page intentionally left blank.

Appendix B Scorecards

EXAMPLE SCORE SHEET

B-1. Tables B-1 and B-2 are examples of proficiency score sheets. This shows all pertinent blocks that need to be filled out appropriately. If additional remarks are needed, a row may be added to accommodate a “Notes” block.

**Table B-1. Leaders Counter small Unmanned Aircraft Systems (C-sUAS) Skills Proficiency
Test Evaluation Scorecard**

<i>Leaders C-sUAS Skills Proficiency Test Evaluation Scorecard</i>			
Leader Name:		Date:	System:
Crew/Section:		Unit:	
Task #	Task Title	Results	Grader's Initials
441-CUAS-0001	Plan Passive Air and Missile Defense (AMD) Measures to Counter Unmanned Aircraft Systems (C-UAS)	Go: No-Go:	
441-CUAS-2001	Plan for Counter small Unmanned Aircraft Systems (C-sUAS) Operations	Go: No-Go:	
441-CUAS-0002	Coordinate with Local Airspace Manager for Unmanned Aircraft Systems Countermeasure Operations	Go: No-Go:	
441-CUAS -1001	Develop a Unit C-sUAS Training Strategy	Go: No-Go:	

**Table B-2. Operators Counter small Unmanned Aircraft Systems (C-sUAS) Skills Proficiency
Test Evaluation Scorecard**

<i>Operators C-sUAS skills proficiency test evaluation scorecard</i>			
Operators Name:		Date:	System:
Crew/Section:		Unit:	
Task #	Task Title	Results	Grader's Initials
441-CUAS-1004	Operate C-sUAS Non-Lethal Systems	Go: No-Go:	
441-CUAS-1006	Operate C-sUAS Detection Devices	Go: No-Go:	
441-CUAS-1002	Operate C-sUAS Lethal Systems	Go: No-Go:	
441-CUAS-1007	Manage Operational Status of C-sUAS Detection Device	Go: No-Go:	
441-CUAS-1005	Manage Operational Status of C-sUAS Non-Lethal Systems	Go: No-Go:	
441-CUAS-1003	Manage Operational Status of C-sUAS Lethal Systems	Go: No-Go:	
071-COM-0028	Load an M16 Series Rifle/M4 Series Carbine	Go: No-Go:	
071-COM-0027	Unload an M16 Series Rifle/M4 Series Carbine	Go: No-Go:	
071-COM-0029	Perform a Function Check on an M16-Series Rifle/M4- Series Carbine	Go: No-Go:	
071-COM-1017	Compute Back Azimuths	Go: No-Go:	
071-326-3003	React to Aerial Contact at Platoon Level	Go: No-Go:	
441-CUAS-1008	Perform Destruction of C-sUAS Equipment	Go: No-Go:	

B-2. Evaluators must keep in mind that even the best engagements can be improved to make the crew more efficient and effective. Crews will have tailored gunnery training based up the score sheet's information and overall gunnery performance. Trainers cannot address key issues if they are not recorded in a standardized and understandable manner.

Appendix C

C-sUAS Written Test Example

- C-1. What frequencies do C-sUAS systems target for disruption?
- 433 MHz
 - 900 MHz
 - 2.4 GHz and 5.8 GHz
 - L1 at 1575.42 MHz
 - L2 at 1227.6 MHz
- C-2. What common optical devices are used and available to detect C-sUAS?
- Night vision goggles are useful for nighttime identification of the C-sUAS. It will also help determine whether the object looked at is a conventional aircraft or a closer UAS.
 - Binoculars can be used in a similar way to look at objects to get a better visual for positive identification.
- C-3. How are search sectors developed for C-sUAS?
- Search Sectors should be determined while keeping in mind established named area of interest (NAI)s that have been developed from previous reporting. launch and recovery points can be established as a part of those named area of interests that correlate to places that the sUAS operators have historically been located.
- C-4. How are Range Cards used for C-sUAS?
- Range cards will be used similarly to how they are for a ground fight. Keep in mind that the altitude of the sUAS can be important, so when dictating terrain features, annotating the height can be useful in estimating sUAS altitude for more accurate reporting.
- C-5. What are some operational considerations when employing handheld C-sUAS platforms?
- Some C-sUAS systems works within line of sight, including when the enemy UAS is above the patrol or guard tower.
 - Once you engage an enemy UAS with the C-sUAS system, YOU MUST maintain sight alignment with the UAS until it either departs the Area of Operations and regains its C2 link or becomes disrupted and lands.
 - Obtain compass bearings of UAS's route of ingress and egress (if it returns to the initial launch point).
 - It is important to note that the target engagement authority in the base defense operations center (known as the BDOC) is the only one to authorize use of any type of jamming on a suspected enemy UAS.
- C-6. What is the difference between Area Defense and Local Defense?
- Area covers a wide array of assets while local is a smaller, point defense.
- C-7. How does Line of Sight affect C-sUAS systems?
- The system must have electronic line of site with the RF signal.
- C-8. When are the C-sUAS Systems detecting and identifying its target?
- When the system is enabled but the system is not jamming, it acts as an RF power meter.
- C-9. What is the proper technique of recovering a Threat UAS?
- If an enemy UAS lands or is grounded, keep the Drone Buster centered on the UAS and utilize the 5 Cs to begin exploitation/recovery efforts.
- C-10. What is Jamming?
- Deliberate interference with wireless communications.
 - Overloading a receiver with a high-power transmission.
 - Same frequency as signal of interest.
 - Same type of modulation.

- Higher power output.
- Increase in noise to signal ratio.

C-11. What are the types of Jamming?

- Spot Jamming:
 - All power focused on a single frequency.
 - Requires lower output.
- Sweep Jamming:
 - Full power shifted from one frequency to another.
 - Lower output due to one frequency jammed at a time.
- Barrage Jamming:
 - Jamming multiple frequencies at one time with a single system.
 - Higher power output required due to the number of frequencies jammed.

C-12. What operating weights and altitudes characterize a group 2 UAS?

- 35lbs; flying 1000ft AGL.
- 45lbs; flying 2500ft AGL.

C-13. Name three categories of UAS.

- Fixed-Wing.
- Rotary-wing.
- Hybrid.

C-14. True or false: There are noticeable effects of GPS disruption on nearby systems.

- True.

C-15. True or False: C2 Jamming can disconnect a UAS' video feed?

- True.

C-16. What do you do after engaging a UAS?

- Send up an UAS engagement report.

C-17. The UAS Detection Report consists of 7 lines. Name all 7 Lines in order of precedence.

- Line 1: Location of observer.
- Line 2: DTG.
- Line 3: Distance.
- Line 4: Direction.
- Line 5 Heading and speed.
- Line 6: Altitude.
- Line 7: UAS Type.

C-18. True or false: If the threat UAS lands within the AO the operator should immediately retrieve the UAS.

- False.

C-19. What is a "COTS" UAS?

- Commercial off the shelf.

C-20. True or false: UASs cannot operate on Wi-Fi?

- False.

C-21. When planning for C-sUAS attacks leaders refer to what ATP?

- ATP 3-01.8 *Techniques for Combined Arms for Air Defense*.

C-22. UAS platforms are grouped based on the physical and performance characteristics of weight, operating altitude, and airspeed. How many DoD UAS groups are there?

- Groups 1-5.

C-23. What level(s) of the force does UAS pose a significant threat?

- All levels of the force.

C-24. When engaging with small arms defense techniques what four techniques must be determined?

- Small arms engagement.
- Volume of fire.
- Football field technique.
- Aiming point.

C-25. At what aiming point are you to engage a Threat UAS when coming in your direction?

- Slightly above UAS body.

C-26. Identify the rules of engagement that apply to all personnel in reference to threat UAS.

- Right of self-defense.
- Hostile identification criteria.
- Weapons control status.

C-27. What are the two initial reaction steps to seeing a threat UAS?

- Determining distance and bearing to the threat.

C-28. What type of report should you use to report sightings of a threat UAS?

- Standardized UAS Detection Report (7 Line).

This page intentionally left blank.

Appendix D

Emerging C-UAS Systems

This appendix aims to provide a foundational overview of the generalized equipment that a unit might encounter or be issued. The goal is to equip service members with the essential knowledge needed to successfully execute a counter C-UAS mission, thereby enhancing the Army's overall readiness and effectiveness in countering the UAS threat. Topics covered will include system components, basic operation, routine maintenance, and safety protocols, ensuring that units are well-prepared to integrate C-UAS capabilities into their operations.

ARMY'S THREAT UAS RESPONSE AND TRAINING NEEDS

D-1. Currently, the Army is actively addressing the increasingly complex and dynamic threat posed by unmanned aircraft systems. In response to this evolving challenge, the Army has initiated the deployment of non-standard and prototype C-UAS equipment to various operational locations around the globe.

D-2. These systems are designed to detect, track, and neutralize potential threats from unmanned aircraft, ensuring the safety and security of military personnel and critical infrastructure.

COMMANDER'S RESPONSIBILITIES

D-3. Commanders must ensure that all tasks are completed in strict accordance with the manufacturer's operator manual, adhering to the specific instructions and guidelines provided for the safe and effective use of the equipment.

D-4. Commanders will enforce the use of appropriate personal protective equipment (PPE) as required, ensuring the safety of all personnel involved in the operation and maintenance of the equipment.

D-5. Commanders must ensure service members follow the unit's SOP, which outline the specific protocols and best practices for conducting tasks within their unit, ensuring consistency, efficiency, and compliance with all relevant regulations and standards.

EMERGING C-UAS EQUIPMENT TASKS

D-6. It is crucial for commanders to guarantee that their service members are adequately trained to operate and maintain the C-UAS equipment they encounter or are issued.

D-7. Effective use of this technology requires a solid understanding of its capabilities, limitations, and operational procedures. However, due to specific contractual agreements for certain C-UAS equipment, some maintenance and operational tasks may need to be performed by designated field support representatives (FSRs). Even in these cases, service members should possess a basic familiarity with all tasks to ensure seamless coordination and efficient troubleshooting.

D-8. Commanders should plan to utilize their C-UAS equipment to engage targets at the range, provided it is permitted by their respective installation.

D-9. Below you will find the fundamental information required to effectively maintain and operate the various types of C-UAS equipment available.

- **C-UAS Command and Control Equipment (C2) Tasks**
 - PMCS (Preventive Maintenance Checks and Services) – Regular checks and services to ensure the C2 equipment is in good working condition. This includes cleaning, lubricating, and inspecting for any signs of wear or damage.
 - Understand and follow all safety guidelines, including manufacturer's instructions and proper training, to ensure safe and effective equipment use.
 - PM&E (Prepare for Movement and Emplacement) – Preparing the C2 equipment for transportation and setting it up in a new location. This includes securing the equipment for travel, unpacking, and installing it at the new site.
 - Power up – Turning on the power to the C2 and ensuring it boots up correctly.

- Initialize – Setting the initial parameters and configurations for the C2 equipment to function properly.
- Diagnostics/BITE (Built-In Test Equipment) – Running tests to identify any faults or issues with the C2 system. BITE are tests built into the system itself.
- Configure System – Setting up the C2 equipment according to the specific mission requirements.
- Establish communication link (data) – Connecting the C2 system to the data communication network to transmit and receive data.
- Demonstrate the knowledge, skills, and attributes to perform airspace deconfliction.
- Battle drills – Conducting practice exercises based on lessons learned from tactical environments to ensure readiness for combat situations.
- Power down – Safely shutting down the equipment and turning off the power.
- **Sensor Tasks**
 - PMCS (Preventive Maintenance Checks and Services) – Regular checks and services to ensure the sensor equipment is in good working condition. This includes cleaning, lubricating, and inspecting for any signs of wear or damage.
 - Understand and follow all safety guidelines, including manufacturer's instructions, proper training, and radiation hazard zones to ensure safe and effective equipment use.
 - PM&E (Prepare for Movement and Emplacement) – Preparing the sensor system for transportation and setting it up in a new location. This includes securing the equipment for travel, unpacking, and installing it at the new site.
 - Power up – Turning on the power to the sensor system and ensuring it boots up correctly.
 - Initialize – Setting the initial parameters and configurations for the sensor system to function properly.
 - Diagnostics/BITE (Built-In Test Equipment) – Running tests to identify any faults or issues with the sensor system. BITE are tests built into the system itself.
 - Configure System – Setting up the sensor system according to the specific mission requirements.
 - Establish communication link (data) – Connecting the sensor system to the data communication network to transmit and receive data.
 - Battle drills – Conducting practice exercises based on lessons learned from tactical environments to ensure readiness for combat situations.
 - Power down – Safely shutting down the sensor system and turning off the power.
- **Electromagnetic Attack (EA) Tasks**
 - PMCS (Preventive Maintenance Checks and Services) – Regular checks and services to ensure the EA system is in good working condition. This includes cleaning, lubricating, and inspecting for any signs of wear or damage.
 - Understand and follow all safety guidelines, including manufacturer's instructions and proper training, to ensure safe and effective equipment use.
 - PM&E (Prepare for Movement and Emplacement) – Preparing the EA system for transportation and setting it up in a new location. This includes securing the equipment for travel, unpacking, and installing it at the new site.
 - Power up – Turning on the power to the EA system and ensuring it boots up correctly.
 - Initialize – Setting the initial parameters and configurations for the sensor system to function properly.
 - Diagnostics/BITE (Built-In Test Equipment) – Running tests to identify any faults or issues with the EA system. BITE are tests built into the system itself.
 - Configure System – Setting up the EA system according to the specific mission requirements.
 - Establish communication link (data) – Connecting the EA system to the data communication network to transmit and receive data.
 - Demonstrate the ability to establish, maintain, and verify EA authority.
 - Demonstrate capability to effectively engage targets that replicate the threat.
 - Battle drills – Conducting practice exercises based on lessons learned from tactical environments to ensure readiness for combat situations.
 - Power down – Safely shutting down the EA system and turning off the power.
- **Interceptor Platform Tasks**

- PMCS (Preventive Maintenance Checks and Services) – Regular checks and services to ensure the interceptor platform is in good working condition. This includes cleaning, lubricating, and inspecting for any signs of wear or damage.
- Understand and follow all safety guidelines, including manufacturer's instructions and proper training, to ensure safe and effective equipment use.
- PM&E (Prepare for Movement and Emplacement) – Preparing the interceptor system for transportation and setting it up in a new location. This includes securing the equipment for travel, unpacking, and installing it at the new site.
- Power up – Turning on the power to the interceptor system and ensuring it boots up correctly.
- Diagnostics/BITE – Running tests to identify any faults or issues with the interceptor system.
- Establish communication link (data) – Connecting the interceptor system to the data communication network to transmit and receive data.
- Battle drills – Conducting practice exercises based on lessons learned from tactical environments to ensure readiness for combat situations.
- Demonstrate capability to effectively engage targets that replicate the threat.
- Demonstrate ability to perform launcher reload.
- Power down – Safely shutting down the interceptor system and turning off the power.
- **Gun/Projectile Firing Tasks**
 - Clearing procedures – Ensuring the gun is safe and clear of any ammunition before handling or performing maintenance.
 - Understand and follow all safety guidelines, including manufacturer's instructions and proper training, to ensure safe and effective equipment use.
 - PMCS (Preventive Maintenance Checks and Services) – Regular checks and services to ensure the gun is in good working condition. This includes cleaning, lubricating, and inspecting for any signs of wear or damage.
 - Disassemble – Taking the gun apart for cleaning, maintenance, or storage.
 - Assemble – Putting the gun back together after cleaning or maintenance.
 - PMI (Preventive Maintenance Inspection) – Inspecting the gun for any signs of wear or damage that could affect its performance.
 - Load – Safely loading ammunition into the gun.
 - Perform immediate action – Taking quick action to correct any malfunctions that occur during the operation of the gun.
 - Zero – Adjusting the sights of the gun to ensure accuracy.
 - Demonstrate capability to effectively engage targets that replicate the threat.
 - Unload – Safely removing ammunition from the gun.

This page intentionally left blank.

Appendix E

Handheld Certification and Qualification

The certification and qualification tables presented in this section are how to develop C-sUAS platform proficiency, execute supporting collective tasks, and provide an objective approach to evaluate, certify, and qualify C-sUAS individual operators and crews. The C-sUAS tables I-VI (table E-1) will use the basic-intermediate-advance method. The basic phase trains individual task proficiency through the execution of C-sUAS Table I and results in individual certification. Individual certification is accomplished through the written test and C-sUAS skills proficiency test (C-sUAS skills proficiency test). The intermediate phase trains individual operators' proficiency through the execution of C-sUAS tables II-V. C-sUAS table I (individual certification) is required to proceed to the follow-on tables II-VI, table IV (individual qualification practice) and table V (individual qualification) are required to proceed to table VI. The advanced phase, table VI (collective qualification) outlined in this chapter are the minimum requirement to meet the Army standards for certification and qualification. C-sUAS operators are required to qualify bi-annually, and evaluators are required to certify on tables I-VI biannually. Tables II-VI must be completed within six weeks of table I.

Table E-1. Counter small Unmanned Aircraft Systems (C-sUAS) Tables I through VI

Table	Description
TI	Individual Written Test Written Safety Test C-sUAS Skills Proficiency Test
T II	Threat Identification/Reporting
T III	Calibrate/Zero (Basic)
T IV	Individual Practice
T V	Individual Qualification
T VI	Collective Qualification

CERTIFICATION TABLES

TABLE I: INDIVIDUAL CERTIFICATION

E-1. Table I consists of three primary requirements. The written test composed of a minimum of 25 questions on C-sUAS platforms and threats. The written test should be administered to all C-sUAS assigned personnel prior to being assessed on the individual proficiency tasks in the C-sUAS skills proficiency test. The leaders identified in the table below must also be safety certified by achieving a passing score on the safety test. The written test, safety test, and the C-sUAS skills proficiency test listed in this chapter provide the commander a means to evaluate service member's C-sUAS system proficiency prior to any live fire event. All assigned C-sUAS operators, crews are required to pass the written test and pass the C-sUAS skills proficiency test prior to moving on to the subsequent C-sUAS tables. See tables E-2, E-3, and E-4 on page 48 for Table I elements.

Table E-2. Table I, Resources Required

Table I, Resources Required	
Ammunition	No ammunition required. Dummy 5.56mm rounds may be utilized for functions checks on the Smart Shooter.
Equipment	Bal Chatri, Modi, Dronebuster, android tablet, Smart Shooter, Graphical Training Aids, Computer with projector, Tables, and chairs for up to 32 service members
Ranges	Unit Training Area, Mission Training Complex (Classroom environment) Four separate stations for evaluation of each operator
Targets	N/A * Static displays for educational purpose optional

Table E-3. Table I, Task, Conditions, and Standards

Table I, Task, Conditions, and Standards	
Task:	Written Test and Counter small Unmanned Aircraft Systems (C-sUAS) skills proficiency test evaluation.
Condition:	Given each assigned C-sUAS system/effector service members will be evaluated on their general knowledge on each system, the aspects and threat of UAS'. Service members must be able to configure each C-sUAS platform into an operational state.
Standard:	Service members must be able to assemble and place into operations each C-sUAS platform in a time to be determined by the Commander or their Master Evaluator. Operators must be able to score at least 80% on the Written Test. Time standards will be established IAW local Standing Operating Procedures (SOP).

Table E-4. Table I, Test Requirements

Table I, Test Requirements			
Position	Written Test	Safety Certification Test	C-sUAS Skills Proficiency Test
Commander	X		
Platoon Leader	X	X	
Platoon Sergeant	X	X	X
Crew/Section Chief	X		X
Individual Operator	X		X

TABLE I CERTIFICATION REQUIREMENTS**Written Test**

E-2. The unit trainer must develop the written test. The written test consists of a minimum of 25 questions from common operator training, system specific training, maintenance, technical procedures, theory of operations, and safety procedures. An example of the written test is provided in Appendix C. Individuals must score at least 80% on the written test to pass. If the individual fails the written test, retraining and retesting will occur three times, and then the service member will be sent back to unit for retraining or removed from the roster.

Safety Certification Test

E-3. Unit and local range regulations will dictate how units implement safety procedures. The battalion commander must ensure that leaders have been trained and evaluated against these regulations as they pertain to live fire operations. If the written test or the local range control training requirements do not evaluate their knowledge of safety, the unit trainer must create a separate test to be referred to as the safety certification test. This test consists of surface danger zones, electromagnetic spectrum management, radio frequency (RF) hazards and any other safety procedures.

C-sUAS Skills Proficiency Test

E-4. All C-sUAS personnel will be administered the C-sUAS skills proficiency test on all platforms issued to the unit. To pass the C-sUAS skills proficiency test, service member must receive a GO on all stations. If a task is failed, the Service member must be retrained and retested on that station until a GO is received. Appropriate operators' manuals and other references listed for each task on the combined arms rehearsal must be used to prepare, administer, and evaluate the C-sUAS skills proficiency test.

E-5. Upon successful completion of the C-sUAS skills proficiency test, the Service member is certified to perform duty position on any C-sUAS platform individually and in any crew /section throughout Tables II-VI. Table E-5 lists the C-sUAS skills proficiency test certification requirements common to all personnel.

Table E-5. Table I, Counter small Unmanned Aircraft Systems (C-sUAS) Skills Proficiency Test Certification Requirements Common to All

<i>C-sUAS Table I, C-sUAS Skills Proficiency Test Certification Requirements Common to All</i>						
<i>Task #</i>	<i>Task Description</i>	<i>Operator</i>	<i>TM Chief SEC SGT</i>	<i>PSG</i>	<i>PL</i>	<i>CDR</i>
441-CUAS-0001	Plan Passive Air and Missile Defense (AMD) Measures to Counter Unmanned Aircraft Systems			X	X	X
441-CUAS-1005	Plan for Counter small Unmanned Aircraft Systems (C-sUAS) Operations			X	X	X
071-COM-0028	Load an M16 Series Rifle/M4 Series Carbine	X	X			
071-COM-0027	Unload an M16 Series Rifle/M4 Series Carbine	X	X			
071-COM-0029	Perform a Function Check on an M16-Series Rifle/M4-Series Carbine	X	X			
441-CUAS-1004	Operate C-sUAS Non-Lethal Systems (1c. Assemble Dronebuster for Operational use)	X	X			
441-CUAS-1006	Calibrate Dronebuster *see Operators Manual	X	X			
441-CUAS-1006	Prepare dismounted Modi system for operations	X	X			
441-CUAS-1002	Operate C-sUAS Lethal Systems (3. Perform Pre-operational Checks on the Smart Shooter)	X	X			

PLANNING CONSIDERATIONS

E-6. The evaluator tasked to conduct the skills proficiency test should consider personnel, equipment, and material during their planning. The list below should be considered for all stations.

- Primary evaluator, (SGT or above) in charge of administering the test.
- Station evaluator, (SGT or above) one per test station.
- Classroom or training area
- Stopwatch (as per timed station).
- Criterion-scoring checklist (one per service member)

C-sUAS Skills Proficiency Test Score Card

E-7. C-sUAS skills proficiency test score card enables a commander to certify individual proficiency of service members and continue to the following C-sUAS Tables. See Appendix B for C-sUAS skills proficiency test Score Cards.

Safety Precautions

E-8. The commander will incorporate risk assessment in planning all aspects of this training. The unit will conduct a safety briefing IAW the local SOP, and local range operations requirements.

TABLE II: THREAT IDENTIFICATION AND REPORTING

E-9. Table II is designed to assess service members proficiency on the procedures for threat identification by group and category of UAS in both day and night environments, and reporting requirements for detection of UAS, (simulated) engagement of UAS, and incidents involving UAS. See tables E-6, E-7, E-8, E-9, on page 50 and E-10 on page 51, for table II elements.

Table E-6. Table II, Resources Required

<i>Table II, Resources Required</i>	
Ammunition:	No ammunition required
Equipment	Graphical Training Aids, Computer with projector.
Ranges:	Unit Training Area, Mission Training Complex (Classroom environment)
Targets:	N/A * Static displays for educational purpose optional

Table E-7. Table II, Task, Conditions, and Standards

<i>Table II, Task, Conditions, and Standards</i>	
Task:	Correctly identify by group and category threat small Unmanned Aircraft Systems (sUAS).
Condition:	Threat identification and reporting drills table II is a live or simulated training environment, hands-on training event utilizing training aids and devices. It is used to train and evaluate the service members ability to execute critical tasks using their assigned weapon/effector and equipment to identify and report threat unmanned aircraft system (UAS) by group and category, during day and night conditions. Table II may also be executed as concurrent training on all live-fire events (Tables IV through VI). Trainers may utilize physical UAS, mock-ups, and or Graphical Training Aids (GTA) to administer table II.
Standard:	Service members must be able to identify, categorize, and discriminate eight out of ten UAS by group and category. Service members must be able to report (simulated) detection, and engagement of UAS utilizing the reports contained in ATP 3-01.81.

Table E-8. Table II, Individual and Collective Tasks

<i>C-sUAS Table II, Individual and Collective Tasks</i>						
<i>Task #</i>	<i>Task Description</i>	<i>Operator</i>	<i>TM Chief SEC SGT</i>	<i>PSG</i>	<i>PL</i>	<i>CDR</i>
441-CUAS-0002	Coordinate with Local Airspace Manager for Unmanned Aircraft Systems Countermeasure Operations		X	X	X	X
44-BTRY-1000	Defend against Low Slow Small Unmanned Aircraft Systems			X	X	X
441-CUAS-2001	Categorize unmanned aircraft system (UAS) by group and type	X	X			
441-CUAS-2001	Report detection of UAS	X	X			
441-CUAS-2001	Report (simulated) Engagement of UAS	X	X			

Table E-9. Performance Measures

<i>Performance Measures</i>			
PERFORMANCE MEASURES	GO	NO-GO	NA
Categorized by type and group 8 of 10 unmanned aircraft system (UAS)			
Reported Detection and Engagement of UAS			

Table E-10. Table II, Threat Identification and Reporting Drills

Table II, Threat Identification/Report Drills						
Event Name	Drills		Environment		Live	1
Effector	N/A		Conditions		Hands on	
Type Training	Foundation		Primary Facility		STT	Days Required
Required	Yes		Alternate Facility		UTA	
			Component	Active Component	Reserve Component	32 SERVICE MEMBERS
External Evaluation	First Line Leader		Frequency	2	1	
Valid For	T+6		Annual Days Total	2	2	THRUPUT
	TABLE					
	I	II	III	IV	V	VI
	C-sUAS skills proficiency test	Threat Identification	Calibrate/Zero	Individual Practice	Individual Qualification	Collective Qualification
		Required				2.5
When executed condemnation Criteria (CC)	T-6 Through T-X or T-X					Training days to certification
	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
	Time	Key Leader Turnover	Commander Assessment	Subordinate element proficiency	Specialty Proficiency	Staff Section Proficiency
Remarks: Each service member assigned a C-sUAS platform must successfully complete Table II, before any live-fire event. Table II should be used as concurrent training on Tables III, IV, V, and VI while Service members are awaiting training within a firing order or between day and night phases.						
STT-sergeant's time training		T+6-training week or month plus 6			T-A-training area	
T-X-training event exercise		UTA-unit training assembly				

TABLE III: CALIBRATE AND ZERO EQUIPMENT

E-10. 2-10. Individual C-sUAS platforms will be calibrated and zeroed using training ammunition, simulated training environments, tracking practice, and live and stationary targets. This table evaluates service members ability to calibrate and zero assigned C-sUAS platforms. This table builds upon those skills trained and tested during tables I, and II. Table III is the first training event where the service member employs their C-sUAS platform or weapon with training ammunition or simulated effects against a prescribed target to a defined standard. See tables E-11, E-12, E-13, E-14, on page 52 and E-15 on page 53, for Table III elements.

Table E-11. Table III, Resources Required

Table III, Resources Required	
Ammunition:	M855 Ball (5.56mm) 10 rounds per service member.
Equipment	Bal Chatri, Modi, Dronebuster, Smartshooter, any future sensor or C-sUAS platform to the unit, shelter (Inclement weather), tables, chairs, electrical power outlets (Charge Station), target stands, binoculars, and radios for RSO/OIC/Safeties. Double sided red/white paddles for range safeties.
Ranges:	25 Meter zero Range, Known Distance (KD) Range. Practitioner Training Facility for Electronic Attack (EA) equipment. (Optimal Range would be a single Multi-Purpose Range that accommodates Electronic Attack and Live Fire)
Targets:	Two per service member 25 Meter zero target-M4 Carbine (or Smart Shooter 25 Meter Zero Target if available), one Tracking/Tracing boards, COTS UAS x4.

Table E-12. Table III, Task, Conditions, and Standards

<i>Table III, Task, Conditions, and Standards</i>	
Task:	Calibrate / Zero each C-sUAS Platform.
Condition:	Properly calibrate or zero your assigned equipment. Table III is a live-environment, hands-on training event utilizing live ammunition, training aids and devices. It is used to train and evaluate the service members ability to execute critical tasks of calibrate or zero and perform pre-operations functions checks on their assigned Counter small Unmanned Aircraft Systems (C-sUAS) platform's operator manuals, technical manuals, and local Standing Operating Procedures. See Appendices for M4 Zero Card.
Standard:	Service members must properly calibrate or zero and validate calibration or zero of individually assigned C-sUAS platform IAW appropriate Individual systems technical manuals, Operator Manuals, and Individual Tasks 441-CUAS-1002 for Lethal platforms, 441-CUAS-1004 for non-Lethal platforms Within the time constraints as defined by the units local Standing Operating Procedure (SOP).

Table E-13. Table III, Individual Tasks

<i>Table III, Individual Tasks</i>						
<i>Task #</i>	<i>Task Description</i>	<i>Operator</i>	<i>TM Chief SEC SGT</i>	<i>PSG</i>	<i>PL</i>	<i>CDR</i>
441-CUAS-1007	Manage Operational Status of Counter small Unmanned Aircraft Systems (C-sUAS) Detection Device	X	X			
441-CUAS-1002	Operate C-sUAS Lethal Systems (e. Perform Zeroing Procedures Smart Shooter)	X				
441-CUAS-1004	Operate C-sUAS Non-Lethal Systems	X				
441-CUAS-1006	Operate C-sUAS Detection Devices	X				
441-CUAS-1005	Manage Operational Status of C-sUAS Non-Lethal Systems	X	X			

Table E-14. Table III, Performance Measures

<i>Table III, Performance Measures</i>			
PERFORMANCE MEASURES	GO	NO-GO	NA
Performed zeroing of Smart Shooter			
Calibrated Dronebuster			
Performed pre-operation checks on Bal Chatri			
Configured Modi for operations			

Table E-15. Table III, Individual Systems Calibrate/Zero Equipment

TABLE III, CALIBRATE/ZERO EQUIPMENT						
Event Name	Drills	Environment		Live	0.5	
Effector	ALL SYSTEMS EQUIPED	Conditions		LIVE-FIRE		
Type Training	Foundation	Primary Facility		LFR	Days Required	
Required	Yes	Alternate Facility		MRF		
		Component	Active Component	Reserve Component	32 Service members	
External Evaluation	COACH	Frequency	2	1		
Valid For	T+X	Annual Days Total	2	0.5	THRUPUT	
	TABLE					
	I	II	III	IV	V	VI
	CSPT	Threat Identification	Calibrate/Zero	Individual Practice	Individual Qualification	Collective Qualification
	PRQ	PRQ	Required TX			2.5
When executed condemnation Criteria (CC)						Training days to certification
	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
	Time	Key Leader Turnover	Commander Assessment	Subordinate element proficiency	Specialty Proficiency	Staff Section Proficiency
Remarks: Each Service member assigned an individual weapon must successfully complete Table IV, Basic, as the event includes live-fire zero of aiming devices.						
MRF- modified record fire		LFR- live fire range		PQR- personnel qualification record		
TX-training exercise		T-X-training event exercise				

TABLE IV: INDIVIDUAL QUALIFICATION PRACTICE

E-11. Table IV is live fire training using ammunition on an authorized live fire facility. This table reinforces the basic skills trained during previous tables. See tables E-16, E-17, E-18, E-19, on page 54 and E-20 on page 55, for Table IV elements.

Table E-16. Table IV, Resources Required

Table IV, Resources Required	
Ammunition:	M855 Ball (5.56mm) 40 rounds per service member **more rounds will be required for Commander's option of ground engagements with Smart Shooter.
Equipment	Bal Chatri, Modi, Dronebuster, Smart Shooter, UAS (Drone) X4, Balloons 15 per Smart Shooter operator, water source, zip ties two per balloon, string 200 feet, radios for communicating between threat UAS Operator, evaluator, RSO's and safeties.
Ranges:	Known Distance (KD) Range. Aerial Gunnery Range, or Practitioner Training Facility for Electronic Attack (EA) equipment. (Optimal Range would be a single Multi-Purpose Range that accommodates Electronic Attack and Live Fire)
Targets:	Silhouette targets (Stationary) 10 per service member, COTS UAS x5.

Table E-17. Table IV, Task, Conditions, and Standards

Table IV, Task, Conditions, and Standards	
Task:	Employ each Counter small Unmanned Aircraft Systems (C-sUAS) platform to defeat threat UAS
Condition:	Employ your C-sUAS platform and detect, identify, discriminate, and engage threat UAS. Upon detection a UAS detection report must be sent to higher headquarters. After engagement by any C-sUAS platform an UAS engagement report must be sent. Table IV is a live-environment, hands-on training event utilizing live ammunition, live UAS, training aids and devices. It is used to train and evaluate the service members ability to execute critical tasks of mitigate, deny, or destroy the UAS threat using their assigned C-sUAS platforms, IAW all C-sUAS platform's Operator Manuals, Technical Manuals, and local Standing Operating Procedures.
Standard:	Service members must properly employ individually assigned C-sUAS platforms to mitigate/defeat 8 out of 10 threat UAS IAW appropriate individual systems technical manuals, and operator manuals. Within the time constraints defined by the units local Standing Operating Procedure (SOP). Threat UAS will be a series of both moving and stationary targets defined by the Master Evaluator and or the unit Commander.

Table E-18. Table IV, Individual Tasks Practice

Table IV, Individual Tasks Practice						
Task #	Task Description	Operator	TM Chief SEC SGT	PSG	PL	CDR
441-CUAS-1007	Manage Operational Status of Counter small Unmanned Aircraft Systems (C-sUAS) Detection Device	X	X			
441-CUAS-1004	Operate C-sUAS Non-Lethal Systems	X				
441-CUAS-1006	Operate C-sUAS Detection Devices	X				
441-CUAS-1005	Manage Operational Status of C-sUAS Non-Lethal Systems	X	X			
441-CUAS-0002	Coordinate with Local Airspace Manager for Unmanned Aircraft Systems Countermeasure Operations		X	X	X	X
441-CUAS-1003	Manage Operational Status of C-sUAS Lethal Systems	X	X			
441-CUAS-1003	Effectively engage 10 stationary ground targets	x				
441-CUAS-1003	Effectively engage 10 aerial targets (UAS)	x				

Table E-19. Table IV, Performance Measures

Table IV, Performance Measures			
PERFORMANCE MEASURES	GO	NO-GO	NA
Detected 8 of 10 UAS (Bal Chatri/Dronebuster)			
Sent 8 of 10 UAS Detection reports			
Sent engagement report for Each engagement of UAS			
Coordinated with ASM for Countermeasure Operations			
Kinetically engaged 8 of 10 stationary ground targets			
Kinetically engaged 8 of 10 UAS targets			
Non-Kinetically engaged 8 of 10 UAS targets			

Table E-20. Table IV, Individual Systems Practice

Table IV, Individual Qualification						
Event Name	Basic		Environment		Live	1.0
Effector	ALL SYSTEMS EQUIPED		Conditions		LIVE-FIRE	
Type Training	Foundation		Primary Facility		Training Area	Days Required
Required	No		Alternate Facility		LFR	
			Component	Active Component	Reserve Component	32 Service members
External Evaluation	COACH		Frequency	2	1	
Valid For	T+X		Annual Days Total	2	1	THRUPUT
	TABLE					
	I	II	III	IV	V	VI
	CSPT	Threat Identification	Calibrate/Zero	Individual Practice	Individual Qualification	Collective Qualification
	PQR	PQR	PQR	Required		1
When executed condemnation Criteria (CC)						Training days to certification
	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
	Time	Key Operator Turnover	Commander Assessment	Subordinate element proficiency	Specialty Proficiency	Staff Section Proficiency
Remarks: Each service member assigned an individual weapon should successfully complete Table V, Practice, to gain experience with and confidence in their we Each service member assigned an individual weapon should successfully complete Table V, Practice, to gain experience with and confidence in their weapon, aiming devices, and skill. **A Practitioners Training Range for EA and Live Fire is optimal						
LFR- live fire range PRQ- personnel qualification record T+X-training event exercise						

TABLE V: INDIVIDUAL QUALIFICATION

E-12. Table V is the standard used to determine the live fire proficiency, using training ammunition, employing C-sUAS platforms. Table V provides the commander an assessment of the service members overall proficiency with employing C-sUAS platforms. See tables E-21, E-22, E-23, E-24, on page 56 and E-25 on page 57 for Table V elements.

Table E-21. Table V, Resources Required

Table V, Resources Required	
Ammunition:	M855 Ball (5.56mm) 40 rounds per service member.
Equipment	Bal Chatri, Modi, Dronebuster, Smart Shooter, unmanned aircraft system (UAS) (Drone) X4, Balloons 15 per Smart Shooter operator, water source, zip ties two per balloon, string 200 feet, radios for communicating between threat UAS Operator, evaluator, range safety officer (RSO's) and safeties.
Ranges:	Known Distance (KD) Range. Aerial Gunnery Range, or Practitioner Training Facility for Electronic Attack (EA) equipment. (Optimal Range would be a single Multi-Purpose Range that accommodates Electronic Attack and Live Fire)
Targets:	Silhouette targets (Stationary) 10 per service member, COTS UAS x5. *Optional Wi-Fi Router and Android Tablet for simulation (GCS to UAS)

Table E-22. Table V, Task, Conditions, and Standards

Table V, Task, Conditions, and Standards	
Task:	Employ each Counter small Unmanned Aircraft Systems (C-sUAS) system/effector to defeat threat UAS
Condition:	Employ your C-sUAS platform and detect, identify, discriminate, and engage threat UAS. Upon detection a UAS Detection report must be sent to higher headquarters. After engagement by any C-sUAS platform a UAS Engagement report must be sent. Table V is a live-environment, hands-on training event utilizing live ammunition, live UAS, training aids and devices. It is used to train and evaluate the service members ability to execute critical tasks of mitigate, deny, or destroy the UAS threat using their assigned C-sUAS platform's operator manuals, technical manuals, and local standing operating procedures.
Standard:	Service members must properly employ individually assigned C-sUAS platform to mitigate 8 out of 10 threat UAS IAW appropriate Individual systems technical manuals, operator manuals. Within the time constraints defined by the units local Standing Operating Procedure (SOP)

Table E-23. Table V, Individual Qualification Practice

Table V, Individual Qualification Practice						
Task #	Task Description	Operator	TM Chief SEC SGT	PSG	PL	CDR
441-CUAS-1007	Manage Operational Status of Counter small Unmanned Aircraft Systems (C-sUAS) Detection Device	X	X			
441-CUAS-1004	Operate C-sUAS Non-Lethal Systems	X				
441-CUAS-1006	Operate C-sUAS Detection Devices	X				
441-CUAS-1005	Manage Operational Status of C-sUAS Non-Lethal Systems	X	X			
441-CUAS-0002	Coordinate with Local Airspace Manager for Unmanned Aircraft Systems Countermeasure Operations		X	X	X	X
441-CUAS-1003	Manage Operational Status of C-sUAS Lethal Systems	X	X			
441-CUAS-1003	Effectively engage 10 stationary ground targets	x				
441-CUAS-1003	Effectively engage 10 aerial targets (sUAS)	x				

Table E-24. Table V, Performance Measures

Table V, Performance Measures			
PERFORMANCE MEASURES	GO	NO-GO	NA
Detected 8 of 10 UAS (Bal Chatri/Dronebuster)			
Sent 8 of 10 UAS detection reports			
Sent engagement report for Each engagement of unmanned aircraft system (UAS)			
Coordinated with ASM for Countermeasure Operations			
Kinetically engaged 8 of 10 stationary ground targets			
Kinetically engaged 8 of 10 UAS targets			
Non-Kinetically engaged 8 of 10 UAS targets			

Table E-25. Table V, Individual Qualification

Table V, Individual Qualification						
Event Name	Individual Qualification		Environment		Live	1
Effector	ALL SYSTEMS EQUIPED		Conditions		LIVE-FIRE	
Type Training	Foundation		Primary Facility		Training Area	Days Required
Required	No		Alternate Facility		Alternate Facility	
			Component	Active Component	Reserve Component	
External Evaluation	COACH		Frequency	2	1	
Valid For	T+X		Annual Days Total	2	1	THRU PUT
	TABLE					
	I	II	III	IV	V	VI
	CSPT	Threat Identification	Calibrate/Zero	Individual Practice	Individual Qualification	Collective Qualification
	PQR	PQR	PQR	Required		1
When executed condemnation Criteria (CC)						Training Days to Certification
	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
	Time	Key Operator Turnover	Commander Assessment	Subordinate Element Proficiency	Specialty Proficiency	Staff Section Proficiency
Remarks: Each Service member assigned an individual weapon should successfully complete Table V, Practice, to gain experience with and confidence in their we Each service member assigned an individual weapon should successfully complete Table V, Practice, to gain experience with and confidence in their weapon, aiming devices, and skill.						
PQR- personnel qualification record						

TABLE VI: COLLECTIVE QUALIFICATION

E-13. 2-13. Table VI is the collective qualification table used to determine the live fire proficiency and coordination of a crew of C-sUAS operators working collaboratively to mitigate a UAS threat. Table VI qualifies crews to employ their C-sUAS platforms as a collective event and reinforces the basic skills trained during previous tables. Table VI will be conducted in two separate events, a non-evaluated event and an evaluated event. See tables E-26, E-27, E-28, E-29, on page 58 and E-30 on page 59 for Table VI elements.

Table E-26. Table VI, Resources Required

Table VI, Resources Required	
Ammunition:	M855 Ball (5.56mm) 80 rounds per service member ** Commanders Option scenarios may require additional ammunition
Equipment	Bal Chatri, Modi, Dronebuster, Smart Shooter, UAS (Drone) x8, Balloons 30 per Smart Shooter operator, water source, zip ties two per balloon, string 400 feet, radios for communicating between threat sUAS Operator, evaluator, and safeties.
C2 Node	Additional personnel to replicate a C2 Higher HQ to receive reports.
Ranges:	Known Distance (KD) Range. Aerial Gunnery Range, or Practitioner Training Facility for Electromagnetic Attack (EA) equipment. (Optimal Range would be a single Multi-Purpose Range that accommodates Electronic Attack and Live Fire).
Targets:	COTS UAS x5. (Balloons).
COTS- commercial off-the-shelf UAS- Unmanned aircraft system	

Table E-27. Table VI, Task, Conditions, and Standards

Table VI, Task, Conditions, and Standards	
Task:	Employ multiple Counter small Unmanned Aircraft Systems (C-sUAS) system/effector in a Combined Arms Defense of threat UAS. Successfully engage aerial targets with two or more C-sUAS system/effectors.
Condition:	Employ the appropriate C-sUAS platforms to detect, identify, decide, and engage threat UAS. Upon detection a UAS detection report must be sent to higher headquarters. After engagement by any by any C-sUAS platform, an engagement report must be sent. Engagements of UAS will be conducted IAW all C-sUAS platform operator manuals, technical manuals, and local standing operating procedures. Downed UAS must be treated as a unexploded ordnance (UXO), at no time will the operators approach a downed UAS.
Standard:	Crews must properly employ individually assigned C-sUAS platforms in conjunction with at least one other C-sUAS platform to mitigate 8 out of 10 threat UAS IAW appropriate technical manuals, operator manuals. Within the time constraints defined by the units local Standing Operating Procedure (SOP). Smart Shooter operators will engage 26 or greater of 40 ground and aerial targets to receive a GO. *See Appendix F for Scenario examples.

Table E-28. Table VI, Individual Qualification

Table VI, Individual Qualification						
Task #	Task Description	Operator	Team Chief SEC SGT	PSG	PL	CDR
441-CUAS-1007	Manage Operational Status of Counter small Unmanned Aircraft Systems (C-sUAS) Detection Device	X	X			
441-CUAS-1004	Operate C-sUAS Non-Lethal Systems	X	X			
441-CUAS-1006	Operate C-sUAS Detection Devices	X	X			
441-CUAS-1005	Manage Operational Status of C-sUAS Non-Lethal Systems	X	X			
441-CUAS-0002	Coordinate with Local Airspace Manager for Unmanned Aircraft Systems Countermeasure Operations		X	X	X	X
441-CUAS-1003	Manage Operational Status of C-sUAS Lethal Systems	X	X			

Table E-29. Table VI, Performance Measures

Table VI, Performance Measures			
PERFORMANCE MEASURES	GO	NO-GO	NA
Detected 8 of 10 UAS (Bal Chatri)			
Sent 8 of 10 UAS Detection reports			
Sent engagement report for Each engagement of UAS			
Coordinated with ASM for Countermeasure Operations			
Kinetically engaged 8 of 10 stationary ground targets			
Kinetically engaged 8 of 10 UAS targets			
Non-Kinetically engaged 8 of 10 UAS targets			

Table E-30. Table VI, Collective Qualification

TABLE VI, Collective Qualification						
Individual Qualification						
Event Name	Collective Qualification	Environment		Live	1	
Effector	All systems equipped	Conditions		Live-Fire		
Type Training	Foundation	Primary Facility		Training Facility	Days Required	
Required	Yes	Alternate Facility		Alternate Facility		
		Component	Active Component	Reserve Component	32 Service member	
External Evaluation	Unit	Frequency	2	1		
Valid For	12 months	Annual Days Total	2	1	THRU PUT	
	TABLE					
	I	II	III	IV	V	VI
	CeSPT	Threat Identification	Calibrate/Zero	Individual Practice	Individual Qualification	Collective Qualification
	PQR	PQR	PQR	PQR	PQR	Required T-X
When executed condemnation Criteria (CC)						
	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
		Key Operator Turnover	Commander Assessment	Subordinate Element Proficiency	Specialty Proficiency	Staff Section Proficiency
Remarks: Each Service member assigned an individual weapon must successfully complete Table VI, Qualification to meet their live-fire proficiency gate. The live-fire proficiency gate is required for any Service member to participate in any collective live-fire event.						
Note: Table VI is the Collective Qualification, this table may be used at the Commanders discretion to Qualify crews, and or Sections. It is the Commanders option to determine scenarios that utilize at a minimum two systems and or effectors to mitigate the UAS threat.						
PQR-personnel qualification record						

This page intentionally left blank.

Glossary

The glossary lists acronyms and terms with Army or joint definitions.

SECTION I – ACRONYMS AND ABBREVIATIONS

C2	command and control
CATS	combined arms training strategy
CBRN	chemical, biological, radiological, and nuclear
C-UAS	counter-unmanned aircraft system
C-sUAS	counter-small unmanned aircraft system
EA	electromagnetic attack
FS-LIDS	fixed site-low, slow, small unmanned aircraft system integrated defeat system
GST	Gunnery Skills Test
IAW	in accordance with
IR	infrared
METL	mission-essential task list
M-LIDS	mobile-low, slow, small unmanned aircraft integrated defeat system
RF	radio frequency
SOP	standing operating procedure
TADSS	training aids, devices, simulators, and simulations
TC	training circular
TRP	target reference point
UAS	unmanned aircraft system

This page intentionally left blank.

References

REQUIRED PUBLICATIONS

These documents must be available to intended users for this publication.

Army Dictionary online. Available at <https://jdeis.js.mil/jdeis/index.jsp?pindex=207>.

DOD Dictionary of Military and Associated Terms. 15 February 2025.

FM 1-02.1. *Operational Terms*. 28 February 2024.

FM 1-02.2. *Military Symbols*. 23 January 2025.

RELATED PUBLICATIONS

These documents are cited in this publication.

ARMY PUBLICATIONS

Unless otherwise indicated, most Army doctrinal publications are available online at <https://armypubs.army.mil/>.

ADP 6-0. *Mission Command: Command and Control of Army Forces*. 31 July 2019.

AR 70-62. *Airworthiness of Aircraft Systems*. 11 May 2016.

ATP 3-01.8. *Techniques for Combined Arms for Air Defense*. 29 July 2016.

ATP 3-01.81. *Counter-Unmanned Aircraft System (C-UAS)*. 11 August 2023.

DA PAM 350-38. *Standards in Weapons Training*. 15 November 2024.

DA Pam 385-63. *Range Safety*. 16 April 2014.

FM 6-27/MCTP 11-10C. *The Commander's Handbook on the Law of Land Warfare*. 07 August 2019.

TB MED 524. *Occupational and Environmental Health: Control of Hazards to Health from Laser Radiation*. 31 January 2006.

TC 3-20.0. *Integrated Weapons Training Strategy (IWTS)*. 18 June 2019.

TC 25-8. *Training Ranges*. 2 November 2021.

OTHER PUBLICATION

CJCSM 3212.02E. *Performing Electromagnetic Attack in the U.S. & Canada for Tests, Training, and Exercises*. 18 August 2022. Limited distribution via SIPR.

DoDI 5030.61. *DoD Airworthiness Policy*. 3 December 2024. Available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/503061p.PDF?ver=a4--jubJTtO3HPR0IUN_cQ%3d%3d

PRESCRIBED FORMS

This section contains no entries.

REFERENCED FORMS

Unless otherwise indicated, DA forms are available on the Army Publishing Directorate (APD) website at <https://armypubs.army.mil/>.

DA Form 1594. *Daily Staff Journal or Duty Officer's Log*.

DA Form 8265. *Common Crew Score Sheet*.

DA Form 8265-1. *Common Crew Roll-Up*.

DA Form 2028. *Recommended Changes to Publications and Blank Forms*.

This page intentionally left blank.

Index

Entries are by paragraph number.

B

Battle drill, 2-102, 2-103, 2-104, 2-105, 2-106, 2-107, 2-108, 2-109, 2-115, D-9

F

FS-LIDS, 2-6, 2-82, 2-86, 2-88, 2-90, 2-91, 2-92, 2-95, 2-96, 2-98, 2-101, 2-102, 2-104, 2-106, 2-111

M

M-LIDS, 2-3, 2-25, 2-30, 2-49, 2-59, 2-68, 2-69

TC 3-01.81

06 May 2025

By Order of the Secretary of the Army:

RANDY A. GEORGE

*General, United States Army
Chief of Staff*

Official:

A handwritten signature in black ink, appearing to read 'Mark F. Averill', written in a cursive style.

MARK F. AVERILL

*Administrative Assistant
to the Secretary of the Army
2512204*

DISTRIBUTION:

Active Army, Army National Guard, and United States Army Reserve. Distributed in electronic media only(EMO).

This page intentionally left blank.

